PUNJAB ENFORCEMENT & REGULATORY AUTHORITY (PERA)

Government Street Lighting Project

FUNCTIONAL REQUIREMENTS DOCUMENT

Digital Store, Site Issuance & Installation Monitoring

From Procurement to Installation — Complete Traceability

Document ID	FRD-GSL-001
Version	1.0 — Draft for review
Date	18 September 2026
Derived from	BRD-GSL-001 v1.0 (17 September 2026) and SLP/BRD/SLIMS/2026/01 v1.0 (17 September 2026)
System	SLIMS — Street Lighting Store, Issuance & Installation Management System
Prepared for	Government Street Lighting Project, Punjab Enforcement & Regulatory Authority
Prepared by	Business Analysis & Systems Team
Classification	Official — for internal circulation

Transparent · Traceable · Efficient · For a Brighter Punjab

Document Control

Version history

Version	Date	Author	Change	Status
1.0	18-Sep-2026	Business Analysis & Systems Team	First issue. Derived from BRD-GSL-001 v1.0 and the SLIMS BRD v1.0; reconciles the two and adds use cases, data specifications, state models, validation messages and test cases.	Draft for review

Approvals

Sign-off confirms that this document correctly states what the system must do, and that it may be used as the basis for design, build and acceptance testing.

Role	Name	Signature	Date
Project Director			
Store / Warehouse Officer (lead)			
Lighting Supervisor (representative)			
Approving Officer			
Project / Monitoring Officer			
System Administrator / Head of IT			
Internal Audit			

Source documents

Ref.	Document	Version / date	Use in this FRD
G	BRD-GSL-001 — Business Requirements Document: Digital Store, Site Issuance & Installation Monitoring	1.0 · 17-Sep-2026	Baseline. Prevails where sources differ.
S	SLP/BRD/SLIMS/2026/01 — SLIMS Business Requirements Document	1.0 · 17-Sep-2026	Detailed elaboration where G is silent.
PF	End-to-End Process Flow — Digital Store, Site Issuance & Installation Monitoring (11-stage swimlane and 10-stage summary)	PERA · Sep-2026	Target process, stages and role lanes.
RB	SLIMS reference build (portal and API)	0.1 · 17-Sep-2026	Gap analysis only (Annexure A).

Conventions

- **shall** — mandatory; **should** — expected unless there is a documented reason not to; **may** — optional.
- **FR-MOD-nnn** functional requirement · **UC-nn** use case · **BR-nn** business rule · **VAL-MOD-nnn** validation message · **RPT-nn** report · **NTF-nn** notification · **TC-nn** test case · **SCR-nn** screen · **D-nn** decision required.
- Priority uses MoSCoW: **M** must have for go-live, **S** should have, **C** could have.
- The **Trace** column cites the source: **G** = BRD-GSL-001 (FR number, business rule or section), **S** = SLIMS BRD requirement or rule.

Contents

Document Control 2
Version history 2
Approvals 2
Source documents 2
Conventions 2
1. Introduction 5
1.1 Purpose 5
1.2 Scope 5
1.3 Audience 5
1.4 Source documents and precedence 5
1.5 How the requirements are written 5
2. Reconciliation of Source Documents 6
3. Solution Overview 7
3.1 Functional modules 7
3.2 Channels and devices 7
3.3 Process stages mapped to functions 7
4. Actors, Roles and Data Scope 9
4.1 Roles 9
4.2 Data scope 9
4.3 Segregation of duties 9
5. Functional Requirements 10
5.1 Master Data (MDM) 10
5.2 Inventory & Stores (INV) 11
5.3 Requisition & Workflow (REQ) 14
5.4 Field Installation (FLD) 14
5.5 Exception Management (EXC) 15
5.6 Reporting & Dashboards (RPT) 16
5.7 Notifications & Alerts (NTF) 17
5.8 Security, Access & Audit (SEC) 17
6. Use Case Specifications 18
6.1 Store and inventory 18
6.2 Requisition and issue 19
6.3 Field installation and validation 21
6.4 Returns, traceability and control 22
7. Screen Inventory and Common Behaviour 24
7.1 Screens 24
7.2 Common behaviour of scanning screens 24
8. Data Specifications 26
8.1 Item / article master 26
8.2 Purchase order reference 26
8.3 Store location master 26
8.4 Site master 27
8.5 QR code payload 27
8.6 Goods receipt 27
8.7 Site requisition 28
8.8 Issue to site 28
8.9 Installation 28
8.10 Out-of-order return 29
9. State Models 30
9.1 Item lifecycle 30
9.2 Requisition status 31
9.3 Installation evidence and review status 32
9.4 Return disposition 32

10. Business Rules 33
11. Validation and Error Messages 35
12. GPS, Evidence and Geofence Specification 37
12.1 Capture 37
12.2 Distance calculation 37
12.3 Status determination 37
12.4 Stamping and integrity 37
12.5 Configurable parameters 38
13. Stock Ledger and Reconciliation 39
13.1 Movement types 39
13.2 Reconciliation 39
14. Notifications and Alerts 40
15. Reports and Dashboards 41
15.1 Report catalogue 41
15.2 Dashboard 41
15.3 Drill-down path 42
16. Access Control Matrix 43
17. Audit Trail 44
17.1 Recorded for every action 44
17.2 Protection and retention 44
18. Interfaces 45
18.1 Service interface 45
18.2 External interfaces 45
19. Non-Functional Requirements with Functional Impact 47
20. Traceability Matrix 48
20.1 BRD-GSL-001 functional requirements 48
20.2 BRD-GSL-001 key business rules 48
21. Functional Acceptance Test Cases 49
22. Open Issues and Decisions Required 51
Annexure A — Gap Analysis Against the SLIMS Reference Build 52
Annexure B — Glossary 54

1. Introduction

1.1 Purpose

This document states what the Street Lighting Store, Issuance & Installation Management System (SLIMS) must do. It turns the business requirements of BRD-GSL-001 into precise, testable functions: who performs each action, what data is captured, which rules the system applies, the states that items and transactions move through, the messages users see, and the reports produced. It is the basis for design, build and user acceptance testing.

1.2 Scope

The FRD covers all eleven stages of the target process, from the purchase order reference to management reporting, across eight functional modules (Section 3). Procurement tendering, supplier commercial evaluation and payment processing are out of scope; PO data is used only as a reference for receipt and traceability (G §2).

1.3 Audience

- The development and test teams, who build and verify the functions.
- The business owners of each stage — Store / Warehouse, Lighting Supervisors and Officers, Approving Officers and Project / Monitoring Officers — who confirm the behaviour.
- Internal Audit, who confirm the controls.

1.4 Source documents and precedence

Two business requirements documents and a process flow describe the same system. BRD-GSL-001 and the PERA process flow are the organisation's baseline and take precedence; the SLIMS BRD supplies detail where BRD-GSL-001 is silent. Every point on which they differ is listed in Section 2 with the position this FRD takes. Items marked with a decision number need sign-off before the affected module is built (Section 22).

1.5 How the requirements are written

Each requirement describes one observable behaviour of the system, carries a priority and traces back to its source. Requirements are grouped by module in Section 5; the use cases in Section 6 show how they combine in a user's task; Sections 8 to 17 specify the data, states, rules, messages, reports and access that the requirements rely on; Section 21 gives the test for each.

2. Reconciliation of Source Documents

The table records each point on which BRD-GSL-001 (with the process flow) and the SLIMS BRD differ, and the position this FRD adopts. The remainder of the document is written to the adopted position.

Topic	BRD-GSL-001 and process flow	SLIMS BRD	Position adopted in this FRD	Decision
Store location hierarchy	Store / Warehouse → Rack → Shelf → Bin	Store ▸ Zone / Section ▸ Rack ▸ Bin	Store / Warehouse → Rack → Shelf → Bin, stock held at bin. Shelf may be left blank for yard stock such as poles.	D-01
User roles	Seven roles: Store / Warehouse Officer, Lighting Supervisor, Lighting Officer, Approving Officer, Project / Monitoring Officer, System Administrator, Management / Dashboard User. Process flow adds Procurement / Admin (Reference).	Store Keeper, Store In-Charge, Lighting Supervisor, Lighting Officer, PD Office / MIS, Procurement, Internal Audit	The eight roles of BRD-GSL-001 and the process flow (Section 4). Store Keeper and Store In-Charge become one Store / Warehouse Officer role; Internal Audit uses the Management / Dashboard role with audit-trail access.	—
Who approves requisitions	Approving Officer, within delegated authority	Store In-Charge, escalating to PD Office by value	Approving Officer, with an approval matrix by value and scope; escalation above the officer's limit.	D-03
Who reviews installation exceptions	“Authorised review”; the process flow places exception review with the Project / Monitoring Officer	Lighting Supervisor verifies in the field	Project / Monitoring Officer resolves. The reviewer may ask the Lighting Supervisor for field verification.	D-02
Evidence status values	Valid, Location Exception, Missing Evidence, Pending Review	Confirmed, Pending Verification, Approved Exception	Evidence status Valid / Location Exception / Missing Evidence, plus review outcome Approved / Re-capture Required / Rejected (Section 9.3).	—
Photograph stamping	Generate a controlled location overlay while preserving the original	Coordinates burned onto the captured image	Keep the original unaltered with a hash, and generate a stamped copy from the recorded metadata. Both are immutable.	—
Receipt beyond PO balance	Blocked unless an authorised exception is recorded	Configurable tolerance percentage	Blocked by default; allowed only through an approved over-receipt exception. The tolerance parameter is retained, default 0 %.	D-04
Return disposition	Process flow: inspect and classify Usable / Repair / Scrap; repairable items repaired and re-issued	Repairable, Warranty, Scrap	Usable → back to stock; Repairable → repair or warranty claim, then re-inward on the same QR; Not repairable → scrap and approved disposal.	—
Item status model	Received → In Store → Issued to Site → Installed / Returned / Out-of-Order	Fifteen detailed statuses	Detailed statuses drive the system; each maps to a BRD-GSL-001 summary status used in reports (Section 9.1).	—
Received vs available	Item becomes available after receipt and put-away	Receipt and binning in one step	A Receiving location holds received items until put-away. Receipt and put-away may be recorded together when the officer bins at the gate.	—
Requisition fields	Priority and justification	Purpose and need-by date	All four: required date, priority, justification / purpose.	D-10
Duplicate submission	Transaction IDs / idempotency required	Not specified	Every posting carries a transaction ID and is accepted once.	—
Stock reconciliation	Opening + Receipts + Returns − Issues − Approved Adjustments = Closing	Implied by the ledger	Formula adopted as a report and a control (Section 13.2).	—
Process stages	Process flow: eleven stages (first diagram) / ten stages (second diagram)	Five phases	The eleven-stage model is used to organise this FRD (Section 3.3).	—

3. Solution Overview

3.1 Functional modules

The modules follow the recommended structure of BRD-GSL-001 §18, with notifications drawn out as a module of their own.

Code	Module	Covers	Source
MDM	Master Data	Item / article, PO reference, store locations, site master, organisational scope, geofence, reason codes, parameters	§5, §18(1)
INV	Inventory & Stores	Receiving, QR scanning, put-away, stock ledger, transfers, adjustments, issue to site, returns and repair	§7, §10, §18(2)
REQ	Requisition & Workflow	Digital requisition, approval, reservation, issue authorisation, status tracking	§7, §18(3)
FLD	Field Installation	Site work list, QR verification at the pole, before / after evidence, GPS capture, offline sync	§9, §18(4)
EXC	Exception Management	GPS mismatch, missing evidence, invalid scans, variances, review and resolution, amendments	§9, §18(5)
RPT	Reporting & Dashboards	Operational, site-wise, item-wise, PO-wise, officer-wise and management reporting, drill-down, traceability	§11, §18(6)
NTF	Notifications & Alerts	Event notifications, threshold alerts, escalation	§13
SEC	Security, Access & Audit	Authentication, role and scope enforcement, audit trail, duplicate control	§12, §14, §18(7)

3.2 Channels and devices

Channel	Users	Used for
Web portal (desktop browser)	Store / Warehouse Officer, Approving Officer, Project / Monitoring Officer, Procurement / Admin, Management, System Administrator	Receipt, put-away, issue, returns, approvals, exception review, masters, reports, administration
Mobile application (Android, responsive)	Lighting Officer, Lighting Supervisor	Work list, QR scan at the pole, camera capture with GPS, installation, acknowledgement, out-of-order report, requisition. Works offline.
Hardware QR scanner (keyboard-wedge)	Store / Warehouse Officer	High-volume scanning at the inward gate and issue counter; the scan field accepts the payload followed by Enter.
Label printer (thermal, networked)	Store / Warehouse Officer, Procurement / Admin	Weather-resistant QR labels at article creation and for controlled re-prints.

3.3 Process stages mapped to functions

The eleven stages of the end-to-end process flow, the actor at each stage, what the system does, and the requirements that specify it.

#	Stage	Primary actor	What the system does	Requirements
1	Procurement / PO	Procurement / Admin (Reference)	Hold the approved PO reference; create articles item-code wise; generate and print QR labels	FR-MDM-001–006, 008–011
2	Goods Receipt	Store / Warehouse Officer	Validate each scanned label against the PO and its balance; record condition; post the GRN	FR-INV-001–009
3	QR Scan & Put-away	Store / Warehouse Officer	Assign each item to Store → Rack → Shelf → Bin and make it available	FR-MDM-012–014; FR-INV-010–012

#	Stage	Primary actor	What the system does	Requirements
4	Store Inventory	System	Keep the ledger, balances, location and history of every item	FR-MDM-007; FR-INV-013–021
5	Digital Requisition	Lighting Supervisor	Capture the requisition against a registered site; show availability	FR-MDM-015–017; FR-REQ-001–008
6	Approval	Approving Officer	Approve or reject within authority; reserve stock; generate the Issue Slip	FR-REQ-009–014
7	Issue to Site	Store / Warehouse Officer	Validate each scan against the approved requisition; move custody to the site; print the gate pass	FR-INV-022–029
8	Installation	Lighting Officer	Confirm the right item at the right site; record the installation	FR-FLD-001–004
9	Evidence & GPS Validation	Lighting Officer · System · Project / Monitoring Officer	Capture geo-tagged evidence; validate it against the geofence; route exceptions for review	FR-FLD-005–014; FR-EXC-001–011
10	Return / Repair	Lighting Officer · Store / Warehouse Officer	Receive returns into quarantine; inspect and classify; repair and re-issue, or scrap	FR-INV-030–039
11	Reporting	Project / Monitoring Officer · Management / Dashboard User	Report from the transaction record; dashboards, drill-down, traceability, notifications	FR-MDM-018–020; FR-RPT-001–010; FR-NTF-001–005; FR-SEC-001–009

4. Actors, Roles and Data Scope

4.1 Roles

Role	Responsibilities in the system	Data scope	Main modules
Procurement / Admin (Reference)	Records the approved PO reference and vendor; creates articles and their QR series; resolves discrepancy notes; raises warranty claims.	All stores	MDM, INV (discrepancy, warranty)
Store / Warehouse Officer	Receives against PO, scans and puts away, manages bins, issues against approved requisitions, receives and classifies returns, raises adjustments and transfers.	Assigned store(s)	INV, MDM (locations)
Lighting Supervisor	Maintains sites in their scope, creates and submits requisitions, tracks status, performs field verification when asked.	Assigned zone / sites	REQ, MDM (sites), EXC (field verification)
Approving Officer	Approves or rejects requisitions within delegated authority; approves over-receipt exceptions, stock adjustments and disposals.	Assigned zone / stores; value limit	REQ, INV (approvals)
Lighting Officer	Acknowledges material at site, installs, captures before / after evidence with GPS, reports out-of-order items.	Assigned sites	FLD, INV (returns)
Project / Monitoring Officer	Monitors progress, validates evidence, reviews and resolves exceptions, approves amendments to installation records.	Assigned zone / project	EXC, RPT
Management / Dashboard User	Reads dashboards, reports and the audit trail. Internal Audit is provisioned with this role.	Assigned scope (may be all)	RPT
System Administrator	Configures masters, roles, permissions, approval matrix, workflows and parameters; manages users and passwords.	System-wide	SEC, MDM (parameters)

4.2 Data scope

- Every user is assigned a **scope** made of up to three parts: an organisational unit (Zone, Division or Ward / UC), a set of stores and a set of sites. An empty part means “all within the organisational unit”.
- A store transaction is visible to a user whose scope includes the store. A site transaction is visible to a user whose scope includes the site. Issue and return transactions involve both and are visible from either side.
- Scope is enforced by the service interface on every read and every write. A request for an out-of-scope record returns “not permitted”, not an empty result, and the attempt is audited.
- Reports and dashboards aggregate only in-scope records (see FR-RPT-008).

4.3 Segregation of duties

Pair of actions	Rule	Requirement
Raise and approve a requisition	The approver must not be the requester.	FR-REQ-011
Capture and review an installation exception	The reviewer must not be the officer who captured the evidence.	FR-EXC-008
Raise and approve a stock adjustment	The approver must not be the officer who raised it.	FR-INV-020
Request and approve an over-receipt exception	The approver must not be the receiver.	FR-INV-004
Classify as scrap and approve disposal	Disposal is approved by the Approving Officer, not the Store / Warehouse Officer.	FR-INV-037

5. Functional Requirements

Requirements are listed by module. Priority: M must, S should, C could. Trace: G = BRD-GSL-001, S = SLIMS BRD.

5.1 Master Data (MDM)

Master data establishes what an item is, which purchase order it came from, where it may be stored and where it may be installed. No transaction is possible until the relevant master records exist and are active.

5.1.1 Item / article master

ID	Requirement	Pri.	Trace
FR-MDM-001	The system shall create articles article-wise, one per purchase order line, with a unique Item Code composed as CATEGORY-SUBCATEGORY-SPECIFICATION-PO-LINE (for example `LUM-LED-090W-PO2611-001`). The PO number and line are part of the code and remain attached to the item for life.	M	G: §5.1 · S: FR-MD-05, BR-003, BR-004
FR-MDM-002	The article master shall hold description, category, sub-category, specification / model, unit of measure, serialised or bulk tracking, pack size, rate, warranty period, expected service life, re-order level, status (Active, Blocked, Obsolete) and PO reference history.	M	G: §5.1 · S: §9.3
FR-MDM-003	The unit of measure of an article shall be locked once any transaction exists against it.	M	S: FR-MD-08, BR-028
FR-MDM-004	A Blocked or Obsolete article shall not be selectable on a requisition or issuable; its existing stock remains visible in all enquiries.	M	G: §5.1

5.1.2 QR identification

ID	Requirement	Pri.	Trace
FR-MDM-005	On article creation the system shall generate one unique QR identifier per serialised unit, or per pack / batch for bulk articles. The payload shall carry the item code, a serial number and a check value derived from both (format in Section 8.5).	M	G: §5.5 · S: FR-MD-06
FR-MDM-006	A QR identifier shall be unique across the whole system and never reused, including after the item is disposed. A scan shall never create a duplicate item record.	M	G: §10, Rule 2 · S: BR-002
FR-MDM-007	Scanning a QR in any function shall retrieve the item record with its current status, current store location or site, PO reference and full transaction history.	M	G: §5.5, FR-14 · S: FR-IC-09
FR-MDM-008	The system shall print label sheets showing the QR symbol, item code, description, PO number and serial. A re-print shall carry the same identifier and be logged against the item with the reason.	M	S: FR-MD-07, FR-MD-14

5.1.3 Purchase order reference

ID	Requirement	Pri.	Trace
FR-MDM-009	The system shall hold approved purchase order references — PO number, vendor, PO date, delivery due date, warranty terms and lines (item description, category, specification, unit, quantity, rate) — entered by Procurement / Admin or received by interface.	M	G: §5.2, FR-01 · S: INT-01
FR-MDM-010	For every PO line the system shall maintain ordered, received and outstanding quantities, updated on each receipt and each reversal.	M	G: §5.2
FR-MDM-011	Procurement / Admin shall be able to close and re-open a PO. A closed PO accepts no further receipts.	S	—

5.1.4 Store location master

ID	Requirement	Pri.	Trace
FR-MDM-012	The system shall maintain store locations in the hierarchy Store / Warehouse → Rack → Shelf → Bin and generate a unique location code for each bin (for example `CEN01-R04-S02-B12`). Stock balance shall be held at bin level.	M	G: §5.3, FR-03 · S: §9.1
FR-MDM-013	Each location shall be typed Receiving, Stock, Quarantine, Repair or Scrap. Only Stock locations hold issuable material. Every store shall have at least one location of each type.	M	G: §7 · S: BR-018
FR-MDM-014	A store or location holding stock shall not be deactivated.	M	S: BR-024

5.1.5 Site master

ID	Requirement	Pri.	Trace
FR-MDM-015	The system shall maintain one record per lighting site with a unique Site ID, name / description, administrative hierarchy (Zone → Division → Ward / UC), site type, latitude and longitude to six decimal places, geofence radius, tolerance, assigned Lighting Supervisor and Lighting Officer, and status (Planned, Active, Completed, On hold).	M	G: §5.4, Rule 10 · S: §9.2
FR-MDM-016	Sites shall be loadable in bulk from a survey file, with every row validated and rejected rows reported with the reason.	S	S: FR-MD-04
FR-MDM-017	Changing the coordinates or geofence of a site that has had material issued to it shall require authorised approval, and the previous values shall be retained in the site history.	M	G: §9 · S: BR-025

5.1.6 Scope, reason codes and parameters

ID	Requirement	Pri.	Trace
FR-MDM-018	The system shall maintain the organisational hierarchy used for data scope (Zone, Division, Ward / UC) and the assignment of users to stores and sites.	M	G: §4
FR-MDM-019	The system shall maintain reason-code lists for requisition rejection, receipt discrepancy, failure and condition, stock adjustment and exception resolution.	M	S: §9.5
FR-MDM-020	Authorised administrators shall configure, without code changes: default geofence radius, tolerance, minimum GPS accuracy, permitted clock drift, return period, reservation validity, acknowledgement period, ageing threshold, evidence size limit and minimum justification length.	M	G: §14 · S: FR-MD-12

5.2 Inventory & Stores (INV)

Inventory & Stores covers every physical movement of material through a store: receipt against a PO, put-away to a bin, the stock ledger, transfers and adjustments, issue to a site and the return of out-of-order items. Each movement is a scanned transaction.

5.2.1 Goods receipt

ID	Requirement	Pri.	Trace
FR-INV-001	The Store / Warehouse Officer shall start a receipt by selecting an open PO. The system shall display each PO line with ordered, received and outstanding quantity.	M	G: FR-01, Rule 1 · S: FR-GR-01
FR-INV-002	Each serialised or controlled item shall be received by scanning its QR. On every scan the system shall validate that the label is genuine, belongs to the selected PO, has not already been received, and that the receipt will not exceed the PO line balance — and show the result immediately.	M	G: FR-01, FR-02, Rule 2 · S: FR-GR-02, FR-GR-03
FR-INV-003	Receipt without a PO reference shall not be possible through the standard receipt function.	M	G: Rule 1

ID	Requirement	Pri.	Trace
FR-INV-004	Receipt beyond the PO line balance shall be blocked unless an over-receipt exception is recorded with a reason and approved by an authorised officer; the exception is linked to the GRN.	M	G: §5.2, Rule 9 · S: BR-014
FR-INV-005	The receiver shall record the physical condition of the consignment and may reject individual items with a reason code and optional photograph. Rejected items do not enter stock.	M	G: §7, §8 · S: FR-GR-05
FR-INV-006	The system shall generate a discrepancy / short-receipt note for any shortage, excess or rejection and route it to Procurement / Admin, who record its resolution.	M	G: FR-12 · S: FR-GR-06
FR-INV-007	The receipt shall capture delivery challan number, invoice number and date, receipt date and receiver, and generate a sequential, non-editable GRN number.	M	G: §8 · S: FR-GR-09, FR-GR-10
FR-INV-008	Partial receipt against a PO line shall be supported; the balance remains open for later receipts.	M	S: FR-GR-11
FR-INV-009	An authorised user may reverse a GRN, with a reason, provided none of its items has moved on. Both the original and the reversal remain visible.	S	S: FR-GR-13, BR-022

5.2.2 Put-away and location

ID	Requirement	Pri.	Trace
FR-INV-010	Received items shall first be held in the store's Receiving location and shall not be available for requisition or issue.	M	G: §7
FR-INV-011	Put-away shall be recorded by scanning the item and the destination bin (Store → Rack → Shelf → Bin). The item becomes available only after successful put-away. Where the officer bins on receipt, receipt and put-away may be recorded in one step.	M	G: FR-03, §7 · S: FR-GR-07, BR-005
FR-INV-012	Internal moves between bins shall be recorded by scanning the item and the new bin; the ledger records both locations.	S	G: §10

5.2.3 Stock ledger and balances

ID	Requirement	Pri.	Trace
FR-INV-013	Every stock movement — receipt, put-away, bin move, transfer, issue, return, re-inward, adjustment and disposal — shall post a ledger entry with item, quantity, from and to location or site, transaction ID, user and timestamp.	M	G: Rule 8, §10 · S: FR-IC-01
FR-INV-014	Each controlled item shall have exactly one current inventory state and one current location or site at any time.	M	G: §10, FR-14
FR-INV-015	The system shall show, for each item code and location: on hand, reserved, available, in quarantine and in transit.	M	G: §11 · S: FR-IC-02
FR-INV-016	The system shall reconcile stock for any store and period as Opening + Receipts + Returns − Issues − Approved Adjustments = Closing (Section 13.2), and report any line that does not balance.	M	G: §10
FR-INV-017	No transaction shall reduce stock at any location below zero.	M	S: BR-021
FR-INV-018	The system shall raise a re-order alert when available stock of an article falls below its re-order level.	S	G: §13 · S: FR-IC-04

5.2.4 Transfers, adjustments and physical verification

ID	Requirement	Pri.	Trace
FR-INV-019	Inter-store transfer shall be performed by scan with a despatch leg and a receipt leg; between the two the stock is shown as in transit.	S	G: §18 · S: FR-IS-12
FR-INV-020	A stock adjustment shall require a reason code and approval by a user other than the one who raised it. Original and adjusted values are retained.	M	G: §10, Rule 9 · S: FR-IC-06

ID	Requirement	Pri.	Trace
FR-INV-021	Physical verification shall be performed by scanning. The system shall produce a variance report against book stock; variances are cleared only through approved adjustments.	S	S: FR-IC-05

5.2.5 Issue to site

ID	Requirement	Pri.	Trace
FR-INV-022	Issue shall be possible only against an approved requisition. There is no direct issue function.	M	G: FR-07, Rule 3 · S: FR-IS-01, BR-006
FR-INV-023	The Store / Warehouse Officer shall issue by scanning each item. On every scan the system shall validate that the requisition is approved and not cancelled or exhausted, the item code is on the requisition, the item is available at the issuing store, and the approved quantity is not exceeded.	M	G: FR-06, FR-07 · S: FR-IS-02, FR-IS-03, BR-010
FR-INV-024	An item already issued, installed, reserved for another requisition, in quarantine or scrapped shall be refused at scan, with its current status and location shown.	M	G: §10, §16 · S: FR-IS-04, BR-011
FR-INV-025	The issue shall identify a registered, active Site ID. On posting, each item's location changes from the store bin to the site and its status to Issued to Site.	M	G: Rule 4, Rule 10 · S: FR-IS-06, BR-012
FR-INV-026	Posting shall generate a sequential Issue Note and Gate Pass recording requisition, site, items, quantity, issuing store, user, date and time, the person collecting and the vehicle.	M	G: §8 · S: FR-IS-07, FR-IS-08
FR-INV-027	Partial issue shall be supported. The requisition shows approved, issued and balance quantity per line and becomes Issued when every line is satisfied; any unused reservation on a satisfied line is released.	M	—
FR-INV-028	The Lighting Officer shall acknowledge receipt of issued material on the mobile app, recording the time and device location. Issues not acknowledged within the configured period are flagged.	S	G: §13 · S: FR-IS-09, FR-IS-10
FR-INV-029	Unused issued material shall be returnable to a Stock location by scan, restoring it to available status.	M	S: FR-IS-11

5.2.6 Out-of-order returns and repair

ID	Requirement	Pri.	Trace
FR-INV-030	The Lighting Officer shall report an installed item as out of order or dismantled by scanning it and selecting a reason and condition, with photographs where the configuration requires them.	M	G: FR-13, §8, Rule 7 · S: FR-RT-01
FR-INV-031	The Store / Warehouse Officer shall receive a returned item by scanning its QR. The system shall display its full history and place it in a Quarantine or Repair location — never directly into issuable stock.	M	G: FR-13, Rule 7 · S: FR-RT-02, FR-RT-03, BR-018
FR-INV-032	After inspection the officer shall classify the returned item as Usable, Repairable or Not repairable, with a reason.	M	G: FR-13 · S: FR-RT-05
FR-INV-033	A Usable item shall be moved by scan to a Stock location and become available again.	M	—
FR-INV-034	Warranty eligibility shall be computed by the system from the installation date and the warranty period carried on the item code, and shown at receipt of the return.	M	S: FR-RT-04, BR-019
FR-INV-035	A Repairable item shall be sent for repair or, where within warranty, for a vendor warranty claim; the claim reference and outcome are recorded.	M	S: FR-RT-06
FR-INV-036	A repaired or replaced item shall be re-inwarded to a Stock location on the same QR identifier, keeping its full history, and may then be issued again.	M	G: §10 · S: FR-RT-07
FR-INV-037	A Not-repairable item shall move to a Scrap location. It may be written out of stock only after disposal approval, which records the approver, date and scrap value.	M	G: Rule 9 · S: FR-RT-08, FR-RT-09, BR-020
FR-INV-038	Items reported out of order at site but not received at a store within the configured period shall be flagged and escalated.	S	G: §13 · S: FR-RT-10, BR-029

ID	Requirement	Pri.	Trace
FR-INV-039	When a failed item is replaced, the replacement installation shall record the item it replaces, preserving the site's installation history.	S	S: FR-RT-12

5.3 Requisition & Workflow (REQ)

A requisition is the only route by which material may leave a store. It is raised digitally for a registered site, approved within delegated authority, and on approval ring-fences the stock it needs.

5.3.1 Requisition

ID	Requirement	Pri.	Trace
FR-REQ-001	The Lighting Supervisor shall create a requisition by selecting a registered Site ID from the site master, the issuing store, item codes and quantities, required date, priority and justification / purpose.	M	G: FR-04, §8 · S: FR-RQ-01, FR-RQ-03
FR-REQ-002	Only sites within the supervisor's authorised scope and in Active or Planned status shall be selectable. Free-text site entry shall not be accepted.	M	G: Rule 4, Rule 10, §4 · S: FR-RQ-02, FR-RQ-04, BR-007, BR-008
FR-REQ-003	While the requisition is entered and again on submission, the system shall show available stock at the issuing store for each line and highlight shortfalls.	M	S: FR-RQ-05
FR-REQ-004	A requisition may be saved as a draft and edited until submitted. After submission it is read-only to the requester.	M	—
FR-REQ-005	On submission the system shall assign a sequential requisition number, set status Submitted, route it to the Approving Officer determined by the approval matrix and notify them.	M	G: FR-05 · S: FR-RQ-07
FR-REQ-006	For lines that exceed available stock the system shall record a purchase indent for Procurement / Admin.	S	S: FR-RQ-06
FR-REQ-007	The requester may cancel a requisition that has not been issued; any reservation is released.	M	G: FR-07 · S: FR-RQ-14
FR-REQ-008	The requester shall see the status of each requisition and, per line, the requested, approved, issued and balance quantities.	M	G: §4

5.3.2 Approval

ID	Requirement	Pri.	Trace
FR-REQ-009	The Approving Officer shall approve in full, approve in part with revised quantities, or reject. A rejection requires a reason.	M	G: FR-05 · S: FR-RQ-08
FR-REQ-010	Approval shall respect delegated authority by requisition value and scope. A requisition above the officer's limit shall escalate to the next level in the approval matrix.	M	G: §4 · S: FR-SE-04, BR-026
FR-REQ-011	No user shall approve a requisition they raised.	M	G: Rule 9 · S: BR-026
FR-REQ-012	On approval the system shall reserve the approved quantity against the requisition so it cannot be issued elsewhere, and generate a numbered Issue Slip.	M	S: FR-RQ-10, FR-RQ-11, BR-009
FR-REQ-013	A rejected requisition shall return to the requester with the reason. It may be revised and resubmitted; the history of every version is retained.	M	G: FR-05 · S: FR-RQ-09
FR-REQ-014	Reservations not issued within the configured validity shall be released automatically and the requester notified.	S	S: FR-RQ-13

5.4 Field Installation (FLD)

Field Installation is performed on the Lighting Officer's mobile device. It confirms that the right item is installed at the right site, and captures the evidence on which the system's validation depends.

5.4.1 Installation transaction

ID	Requirement	Pri.	Trace
FR-FLD-001	The Lighting Officer shall see on the mobile app the items issued to the sites assigned to them, grouped by site and issue.	M	G: §4 · S: FR-IN-01
FR-FLD-002	The officer shall scan the item's QR at the installation point. The system shall confirm that the item was issued to that site and is not already installed.	M	G: FR-08, §18 · S: FR-IN-05
FR-FLD-003	The installation transaction shall record issue number, site, item / QR, installed quantity, pole or point identifier, officer, installation date and time, before photograph, after photograph, coordinates and validation status.	M	G: FR-08, §8 · S: FR-IN-11
FR-FLD-004	An installation shall be submittable only when both photographs exist. If a photograph is missing or fails its integrity check after synchronisation, the installation takes evidence status Missing Evidence and is not completed.	M	G: FR-09, Rule 5 · S: BR-013

5.4.2 Evidence capture

ID	Requirement	Pri.	Trace
FR-FLD-005	A BEFORE photograph shall be captured through the app's camera before installation is recorded.	M	G: FR-09, Rule 5 · S: FR-IN-02
FR-FLD-006	An AFTER photograph shall be captured after installation.	M	G: FR-09, Rule 5 · S: FR-IN-06
FR-FLD-007	Photographs shall be captured live through the app. Selecting from the gallery or uploading an existing file shall not be possible.	M	G: §9 · S: FR-IN-03, BR-015
FR-FLD-008	For each photograph the app shall record device latitude, longitude, accuracy and timestamp at the moment of capture. These values shall not be enterable or editable by the user.	M	G: FR-10 · S: FR-IN-04, BR-016
FR-FLD-009	The system shall preserve the original photograph unaltered and generate a controlled stamped copy that overlays date and time, coordinates, accuracy, Site ID, item code and officer (Section 12.4).	M	G: §9 · S: FR-IN-04
FR-FLD-010	The app shall show the current GPS accuracy and the distance to the site point, warn when accuracy is poorer than the configured minimum, and refuse capture beyond three times that minimum.	S	S: §13.1
FR-FLD-011	The system shall refuse evidence from a device reporting a mock location provider or a clock drift beyond the configured limit, and log the attempt.	M	S: FR-IN-13

5.4.3 Offline working and integrity

ID	Requirement	Pri.	Trace
FR-FLD-012	Capture and scanning shall work without connectivity. Transactions queue on the device with their original capture time and coordinates and synchronise automatically when connectivity returns; the synchronisation time is recorded separately.	M	G: §14 · S: FR-IN-14, BR-027
FR-FLD-013	Every field submission shall carry a client transaction ID, so that a retried or repeated synchronisation cannot create a second installation.	M	G: §14
FR-FLD-014	Submitted photographs, coordinates and completed installation records shall not be editable. Corrections use the authorised amendment mechanism in Section 5.5.	M	G: §12 · S: FR-IN-12

5.5 Exception Management (EXC)

The system, not the officer, decides whether evidence is acceptable. Anything it cannot accept becomes an exception that a Project / Monitoring Officer must resolve in writing. Nothing is closed silently.

5.5.1 Validation

ID	Requirement	Pri.	Trace
FR-EXC-001	On receipt of an installation the system shall calculate the distance from the site coordinates to the coordinates of each photograph and compare the larger with the site geofence plus tolerance (Section 12.2).	M	G: FR-11, §9 · S: FR-IN-07
FR-EXC-002	The system shall assign an evidence status of Valid, Location Exception or Missing Evidence, and place every non-Valid installation in Pending Review (Section 12.3).	M	G: §9
FR-EXC-003	A GPS mismatch shall never mark an installation complete. It creates an exception requiring authorised review.	M	G: §9, Rule 6 · S: FR-IN-09, BR-017
FR-EXC-004	The exception queue shall hold GPS mismatch, missing evidence, duplicate or invalid QR scans, quantity variance, stock variance, damaged or out-of-order items and overdue returns or acknowledgements.	M	G: FR-12, §18

5.5.2 Review and resolution

ID	Requirement	Pri.	Trace
FR-EXC-005	The Project / Monitoring Officer shall see an exception queue limited to their scope, filterable by type, site, officer, age and status, showing the measured deviation, the site point and both photographs side by side.	M	G: §4, FR-12
FR-EXC-006	The reviewer shall resolve an exception as Approved (with a written justification of at least the configured length), Re-capture Required (the task returns to the officer) or Rejected (the installation is void and the item remains at site for action).	M	G: Rule 9 · S: FR-IN-10
FR-EXC-007	The reviewer may request field verification by the Lighting Supervisor; the supervisor's findings and photographs are recorded on the exception.	S	S: FR-IN-10
FR-EXC-008	No user shall resolve an exception on a transaction they created.	M	G: Rule 9 · S: BR-026
FR-EXC-009	When an exception is approved the installation is marked complete and the item status becomes Installed. The justification is stored permanently and appears in the exception report.	M	G: Rule 9 · S: BR-017
FR-EXC-010	Unresolved exceptions shall age and escalate according to the notification rules.	S	G: §13
FR-EXC-011	Corrections to completed records shall be made through an amendment request with reason, approval and before / after values. The original record remains visible.	M	G: §12 · S: BR-022

5.6 Reporting & Dashboards (RPT)

Every figure in a report is computed from the transaction record. There is no manual compilation step and therefore no second version of the truth.

ID	Requirement	Pri.	Trace
FR-RPT-001	All reports shall be generated from transaction data without manual compilation.	M	G: FR-15 · S: FR-RP-01
FR-RPT-002	The report catalogue shall include at least the reports in Section 15.1.	M	G: FR-15 · S: FR-RP-02
FR-RPT-003	Reports shall filter by PO, item code, category, store, location, site, administrative area, requisition, officer, date range, status and exception type.	M	G: §11 · S: FR-RP-02
FR-RPT-004	Dashboards shall show stock received, available stock, issued to site, installed, pending installation, GPS-validated installations, GPS exceptions, out-of-order returns, site-wise progress, item-wise consumption, PO utilisation and the transaction audit trail.	M	G: §11 · S: FR-RP-04
FR-RPT-005	Drill-down shall follow Summary → Administrative area → Site → Requisition / Issue → Item → Evidence / validation record, within the viewer's scope.	M	G: §11 · S: FR-RP-05
FR-RPT-006	Any QR or item code shall be traceable in a single view from PO → GRN → store location → requisition → issue → installation evidence → current status.	M	G: §16, FR-14 · S: FR-IC-09

ID	Requirement	Pri.	Trace
FR-RPT-007	Every report shall be exportable to Excel and PDF, with the applied filters recorded in the export.	M	S: FR-RP-03
FR-RPT-008	Report and dashboard content shall be limited to the viewer's scope.	M	G: §11, §12 · S: FR-SE-03
FR-RPT-009	Installations and pending sites shall be viewable on a map, filterable by zone, site and period.	S	S: FR-RP-06, FR-IN-16
FR-RPT-010	Reports may be scheduled for automatic delivery to nominated recipients.	C	S: FR-RP-07

5.7 Notifications & Alerts (NTF)

Notifications move work to the next person without anyone chasing it. Alerts surface patterns that no single transaction reveals.

ID	Requirement	Pri.	Trace
FR-NTF-001	The system shall send the notifications listed in Section 14 to the recipients shown.	M	G: §13 · S: FR-SE-08
FR-NTF-002	Notifications shall be delivered in-app and, where configured, by e-mail and SMS.	M	G: §13 · S: INT-06
FR-NTF-003	Configurable alerts shall cover ageing and turnaround, stock thresholds, repeated GPS mismatches by officer or site, unusual return rates and unresolved site installations.	S	G: §13
FR-NTF-004	An item not actioned within its configured period shall escalate to the next level.	S	G: §13 · S: FR-SE-04
FR-NTF-005	Users shall not be able to switch off notifications of work assigned to them (approvals, exceptions, re-captures).	S	—

5.8 Security, Access & Audit (SEC)

Access is enforced at the service interface, not by hiding buttons. Every action is attributable to one named person.

ID	Requirement	Pri.	Trace
FR-SEC-001	Each user shall authenticate individually with user ID or official e-mail and password. Shared accounts shall not be permitted.	M	G: §14 · S: FR-SE-01
FR-SEC-002	Access shall be role-based as set out in Section 16 and enforced at the service interface as well as the user interface.	M	G: §4, §12 · S: FR-SE-02
FR-SEC-003	Users shall only enter, view, approve or update transactions within their assigned organisational, store and site scope.	M	G: §4, §16 · S: FR-SE-03
FR-SEC-004	Password policy, lock-out after repeated failures, session time-out and forced change of administrator-issued passwords shall apply.	M	G: §14 · S: NFR-12
FR-SEC-005	Every create, submit, approve, reject, issue, return, update and exception-review action shall record user, role, date and time, transaction ID, client address and before / after values.	M	G: §12 · S: FR-SE-05
FR-SEC-006	The audit trail shall be append-only. No posted transaction may be deleted; corrections are made by reversal or amendment.	M	G: §12 · S: FR-SE-06, BR-022
FR-SEC-007	The system shall reject a second posting of any transaction ID, so that double-clicks, retries and repeated synchronisation cannot duplicate a transaction.	M	G: §14
FR-SEC-008	The System Administrator shall configure masters, roles, permissions, workflows and parameters. Every change is audited.	M	G: §4 · S: FR-SE-09
FR-SEC-009	A field device may be bound to a named user; a change of device requires re-authorisation.	S	S: FR-SE-07

6. Use Case Specifications

Each use case shows how the requirements combine in one user task. Validation message IDs refer to Section 11.

6.1 Store and inventory

UC-01 Create article and generate QR labels

Primary actor	Procurement / Admin (Reference); Store / Warehouse Officer
Trigger	An approved PO has been recorded and material is expected.
Preconditions	• The PO is Open. • The PO line has no article yet.
Main flow	1. The user selects the PO and the line. 2. The system proposes the item code from category, sub-category, specification, PO number and line number. 3. The user chooses serialised or bulk tracking, the pack size for bulk, the re-order level and expected life. 4. The system shows how many labels will be generated. 5. The user confirms. 6. The system creates the article, generates the QR series (status Created) and writes the audit entry. 7. The user prints the label sheet.
Alternate and exception flows	• **A1** The line already has an article — the system refuses (VAL-MDM-001). • **A2** The label count exceeds the configured limit — the system refuses and asks for a larger pack size (VAL-MDM-002). • **A3** A label is damaged later — the user re-prints it with a reason; the identifier does not change.
Postconditions	The article is Active; every label exists as an item in status Created, linked to the PO line.
Requirements	FR-MDM-001–002, 005–006, 008

UC-02 Receive goods against a PO by QR scan

Primary actor	Store / Warehouse Officer
Trigger	A vendor consignment arrives at the store with a delivery challan and invoice.
Preconditions	• The PO is Open and its articles and labels exist. • The officer's scope includes the receiving store.
Main flow	1. The officer selects the receiving store and the PO. 2. The system lists the PO lines with ordered, received and outstanding quantity. 3. The officer scans each label. The system validates every scan at once and adds accepted items to the receipt list. 4. The officer records challan number, invoice number and date, and the condition of the consignment. 5. The officer either chooses the destination bin (receive and put away) or leaves the items in Receiving. 6. The officer posts the receipt. 7. The system assigns the GRN number, sets each item to Received (or In Store if binned), updates the PO balance and posts ledger entries.
Alternate and exception flows	• **A1** Forged or unregistered label — refused (VAL-INV-003, VAL-INV-004). • **A2** Label from another PO — refused (VAL-INV-005). • **A3** Label already received — refused (VAL-INV-006). • **A4** Scan would exceed the PO balance — refused (VAL-INV-007); the officer may request an over-receipt exception, which an Approving Officer must approve before the item can be received. • **A5** Damaged or short — the officer rejects the items with a reason; a discrepancy note is created

	for Procurement / Admin.
Postconditions	A GRN exists; accepted items are in the store; the PO balance is reduced; discrepancies are routed.
Requirements	FR-INV-001–008

UC-03 Put away and move stock between bins

Primary actor	Store / Warehouse Officer
Trigger	Items are in the Receiving location, or stock must be re-organised.
Preconditions	• The items are in the officer's store.
Main flow	1. The officer scans the destination bin. 2. The officer scans each item. 3. The system checks the bin is an active Stock location in the same store and moves each item. 4. The system sets received items to In Store and posts ledger entries showing both locations.
Alternate and exception flows	• **A1** Bin is Quarantine, Repair, Scrap or inactive — refused (VAL-INV-009). • **A2** Item is reserved — the move is allowed; the reservation follows the item.
Postconditions	Items are available for requisition and issue at their new bin.
Requirements	FR-INV-010–013; FR-MDM-012

UC-04 Adjust stock after physical verification

Primary actor	Store / Warehouse Officer; Approving Officer
Trigger	Scheduled physical verification, or a variance is discovered.
Preconditions	• A verification count has been opened for the store or location.
Main flow	1. The officer scans every item found at each bin. 2. The system compares the count with book stock and produces the variance report. 3. For each variance the officer raises an adjustment with a reason code. 4. The Approving Officer approves or rejects each adjustment. 5. The system posts approved adjustments to the ledger and closes the count.
Alternate and exception flows	• **A1** The officer attempts to approve their own adjustment — refused (VAL-SEC-003). • **A2** An adjustment would make stock negative — refused (VAL-INV-010).
Postconditions	Book stock equals physical stock; every difference has a reason and an approver.
Requirements	FR-INV-016–017, 020–021

6.2 Requisition and issue

UC-05 Create and submit a site requisition

Primary actor	Lighting Supervisor
Trigger	Work is planned at a site.
Preconditions	• The site exists in the site master, is within the supervisor's scope and is Active or Planned.
Main flow	1. The supervisor chooses the site from the site master and the issuing store. 2. The supervisor adds item codes and quantities; the system shows available stock per line. 3. The supervisor enters the required date, priority and justification. 4. The supervisor submits. 5. The system numbers the requisition, sets it to Submitted, records purchase indents for any shortfall and notifies the Approving Officer.

Alternate and exception flows	• **A1** Site out of scope or inactive — not selectable (VAL-REQ-002, VAL-REQ-003). • **A2** Shortfall on a line — warning only; submission continues (VAL-REQ-004). • **A3** The supervisor saves a draft and submits later.
Postconditions	A submitted requisition awaits approval.
Requirements	FR-REQ-001–006, 008

UC-06 Approve or reject a requisition

Primary actor	Approving Officer
Trigger	A requisition is submitted to the officer.
Preconditions	• The requisition is Submitted and within the officer's scope and value limit.
Main flow	1. The officer opens the requisition and reviews site, lines, availability and justification. 2. The officer approves in full, approves in part by revising line quantities, or rejects with a reason. 3. On approval the system reserves the approved quantity, generates the Issue Slip and notifies the supervisor and the issuing store. 4. On rejection the system returns the requisition to the supervisor with the reason.
Alternate and exception flows	• **A1** Officer raised the requisition — refused (VAL-REQ-005). • **A2** Value above the officer's limit — escalated to the next level (VAL-REQ-007). • **A3** Reject without a reason — refused (VAL-REQ-006). • **A4** Insufficient stock to reserve — the officer approves what is available; the balance remains a shortfall.
Postconditions	Stock is ring-fenced for the requisition, or the supervisor has the reason for rejection.
Requirements	FR-REQ-009–013

UC-07 Issue items to a site by QR scan

Primary actor	Store / Warehouse Officer
Trigger	An approved requisition is ready to be fulfilled.
Preconditions	• The requisition is Approved or Partly Issued. • The officer's scope includes the issuing store.
Main flow	1. The officer selects the requisition; the system shows each line with approved, issued and balance quantity. 2. The officer scans each item handed over; the system validates every scan. 3. The officer records the person collecting and the vehicle. 4. The officer posts the issue. 5. The system numbers the Issue Note and Gate Pass, moves each item from its bin to the site with status Issued to Site, updates the requisition and releases unused reservations on satisfied lines. 6. The officer prints the gate pass.
Alternate and exception flows	• **A1** Requisition not approved, cancelled or exhausted — refused (VAL-INV-020). • **A2** Item not on the requisition — refused (VAL-INV-021). • **A3** Scan exceeds the approved quantity — refused (VAL-INV-022). • **A4** Item already issued, installed, in quarantine or reserved elsewhere — refused (VAL-INV-023, VAL-INV-024). • **A5** Item at another store — refused (VAL-INV-025).
Postconditions	Custody of the items has passed to the site; the store bins are relieved.
Requirements	FR-INV-022–027

UC-08 Acknowledge material at site

Primary actor	Lighting Officer
Trigger	Material arrives at the site.
Preconditions	• The issue is to a site assigned to the officer.
Main flow	1. The officer opens the issue on the mobile app. 2. The officer confirms receipt; the app records time and device location. 3. The system sets the items to At Site.
Alternate and exception flows	• **A1** Not acknowledged within the configured period — the issue is flagged and the supervisor notified.
Postconditions	The items are recorded as physically at the site.
Requirements	FR-INV-028; FR-FLD-001

6.3 Field installation and validation

UC-09 Record an installation with before and after evidence

Primary actor	Lighting Officer
Trigger	The officer is ready to install an item at a site.
Preconditions	• The item is Issued to Site or At Site for a site assigned to the officer. • Location services and the camera are enabled.
Main flow	1. The officer opens the site's work list and scans the item's QR at the pole. 2. The system confirms the item belongs to this site. 3. The officer captures the BEFORE photograph; the app records coordinates, accuracy and time. 4. The officer installs the fixture and enters the pole / point identifier. 5. The officer captures the AFTER photograph; the app records coordinates, accuracy and time. 6. The app shows the distance to the site point. 7. The officer submits. The app sends the transaction with its client transaction ID, both originals and their metadata.
Alternate and exception flows	• **A1** No connectivity — the transaction is queued and synchronised later with its original times and coordinates. • **A2** Accuracy poorer than the standard — warning; capture refused beyond three times the standard (VAL-FLD-005). • **A3** Mock location or clock drift — refused and logged (VAL-FLD-006). • **A4** A photograph is missing — the app does not submit (VAL-FLD-003). • **A5** The same transaction is received twice — the second is ignored (VAL-FLD-007).
Postconditions	An installation transaction exists and is passed to validation (UC-10).
Requirements	FR-FLD-001–008, 010–013

UC-10 Validate evidence and location

Primary actor	System
Trigger	An installation transaction is received.
Preconditions	• The site has coordinates and a geofence.
Main flow	1. The system verifies both photographs are present and intact. 2. It stores the originals, records their hashes and generates the stamped copies. 3. It calculates the distance from the site point to each photograph and compares the larger with geofence + tolerance. 4. Within the limit: evidence status Valid, installation Complete, item Installed. 5. Outside the limit: evidence status Location Exception, review status Pending Review, item

	Pending Review; the Project / Monitoring Officer is notified.
Alternate and exception flows	• **A1** Photograph missing or corrupt — evidence status Missing Evidence, Pending Review; the officer is asked to re-capture.
Postconditions	The installation is either complete or in the exception queue — never complete by default.
Requirements	FR-EXC-001–003; FR-FLD-004, 009

UC-11 Review an installation exception

Primary actor	Project / Monitoring Officer
Trigger	An installation is Pending Review.
Preconditions	• The exception is within the officer's scope and was not created by them.
Main flow	1. The officer opens the exception and sees the deviation, site point, both photographs and their metadata. 2. Optionally the officer requests field verification from the Lighting Supervisor and waits for the findings. 3. The officer resolves: Approved with a written justification, Re-capture Required, or Rejected. 4. On Approved the installation becomes Complete and the item Installed; on Re-capture the task returns to the Lighting Officer; on Rejected the installation is void and the item remains at site.
Alternate and exception flows	• **A1** Justification shorter than the configured minimum — refused (VAL-EXC-001). • **A2** Reviewer created the transaction — refused (VAL-EXC-002).
Postconditions	The exception is closed with a recorded decision; the justification appears in the exception report.
Requirements	FR-EXC-005–009

6.4 Returns, traceability and control

UC-12 Report an out-of-order item at site

Primary actor	Lighting Officer
Trigger	An installed item fails, or is removed during replacement.
Preconditions	• The item is Installed at a site assigned to the officer.
Main flow	1. The officer scans the item. 2. The officer selects Failed in service or Dismantled, the reason and the condition, and captures photographs where required. 3. The system records the report, sets the item to Out of Order or Dismantled, and notifies the receiving store.
Alternate and exception flows	• **A1** Reason or condition missing — refused (VAL-INV-030). • **A2** Item not installed — refused (VAL-INV-031).
Postconditions	The failure is recorded against the site; the item must be returned within the configured period.
Requirements	FR-INV-030, 038

UC-13 Receive, inspect and classify a returned item

Primary actor	Store / Warehouse Officer
Trigger	A removed item is brought to the store.
Preconditions	• The officer's scope includes the store.
Main flow	1. The officer scans the item; the system shows its full history and computed warranty status. 2. The system places it in the Quarantine location and posts the return.

	3. After inspection the officer classifies it Usable, Repairable or Not repairable, with a reason. 4. Usable: the officer scans it to a Stock bin; it becomes available. 5. Repairable: the item goes to repair or, within warranty, to a vendor claim recorded by Procurement / Admin. 6. Not repairable: the item moves to the Scrap location and awaits disposal approval (UC-14).
Alternate and exception flows	• **A1** Item brought back without a field report — the return is recorded with "returned without field report". • **A2** Warranty route chosen for an expired warranty — refused (VAL-INV-032).
Postconditions	The item's disposition is recorded and it sits in the correct location.
Requirements	FR-INV-031–035

UC-14 Re-issue a repaired item, or dispose of scrap

Primary actor	Store / Warehouse Officer; Approving Officer
Trigger	A repair or warranty claim is concluded, or scrap awaits disposal.
Preconditions	• The item is Under Repair, Warranty Claim or Scrap.
Main flow	1. Repaired or replaced: the officer scans the item to a Stock bin; it is re-inwarded on the same QR with its history intact and becomes available. 2. Scrap: the officer submits for disposal with the scrap value; the Approving Officer approves; the item is written out of stock and recorded in the scrap register.
Alternate and exception flows	• **A1** Disposal attempted for an item not classified Not repairable — refused (VAL-INV-033).
Postconditions	The item is back in stock or closed as Disposed.
Requirements	FR-INV-036–037

UC-15 Trace an item end to end

Primary actor	Project / Monitoring Officer; Management / Dashboard User
Trigger	An enquiry, audit or dispute about a specific item.
Preconditions	• The item is within the user's scope.
Main flow	1. The user scans or searches the QR or item code. 2. The system shows the item's PO and vendor, GRN, every location it has occupied, the requisition and issue, the installation with both photographs and validation result, any return and disposition, and its current status. 3. The user opens any linked document or photograph.
Alternate and exception flows	None.
Postconditions	The item's full chain of custody is visible in one view.
Requirements	FR-RPT-006; FR-MDM-007; FR-INV-014

7. Screen Inventory and Common Behaviour

7.1 Screens

ID	Screen	Primary users	Key actions	Channel
SCR-01	Sign in / change password	All	Sign in, first-time password change	Web, Mobile
SCR-02	Operations dashboard	All (scope-filtered)	KPIs, trends, exceptions, pending approvals, overdue returns; drill-down	Web
SCR-03	Purchase orders & vendors	Procurement / Admin	Record PO and lines, open / close, receipt balance	Web
SCR-04	Articles & QR labels	Procurement / Admin, Store / Warehouse Officer	Create article from PO line, generate and print labels, re-print	Web
SCR-05	Goods receipt	Store / Warehouse Officer	Select PO, scan, condition, challan / invoice, post GRN, discrepancy note	Web + scanner
SCR-06	Put-away & bin moves	Store / Warehouse Officer	Scan bin, scan items, move	Web + scanner
SCR-07	Stock position & item enquiry	Store, Monitoring, Management	Stock by bin, reserved / available, ageing, item history	Web
SCR-08	Transfers, adjustments & verification	Store / Warehouse Officer, Approving Officer	Transfer legs, count, variance, adjustment approval	Web + scanner
SCR-09	Requisitions	Lighting Supervisor, Approving Officer	Create, submit, cancel, approve / reject, status	Web, Mobile
SCR-10	Issue to site	Store / Warehouse Officer	Select approved requisition, scan, post, print gate pass	Web + scanner
SCR-11	Field work list	Lighting Officer	Issues by site, acknowledge, start installation	Mobile
SCR-12	Installation capture	Lighting Officer	Scan at pole, before / after capture with GPS, submit, offline queue	Mobile
SCR-13	Exception queue & review	Project / Monitoring Officer, Lighting Supervisor	Review evidence, request field verification, approve / re-capture / reject	Web
SCR-14	Out-of-order report	Lighting Officer	Scan, reason, condition, photographs	Mobile
SCR-15	Returns & disposition	Store / Warehouse Officer, Approving Officer, Procurement / Admin	Receive, classify, repair / warranty, re-inward, disposal approval	Web + scanner
SCR-16	Site master	Lighting Supervisor, System Administrator	Create / edit site, coordinates from device, bulk upload	Web, Mobile
SCR-17	Stores & bins	Store / Warehouse Officer, System Administrator	Create stores and bins, location types	Web
SCR-18	Reports	All (scope-filtered)	Run, filter, drill down, export	Web
SCR-19	Audit trail	Monitoring, Management, System Administrator	Search and view audit entries	Web
SCR-20	Users, roles & parameters	System Administrator	Users, scope, passwords, approval matrix, parameters	Web

7.2 Common behaviour of scanning screens

- Every scan field keeps the keyboard focus, accepts the payload followed by Enter from a hardware scanner or the camera, and validates on the server before the item joins the list. The result — accepted, or refused with the message and rule reference — is shown beside the field.
- A refused scan never enters the list, so a posting can only contain items that have already passed validation; the posting is validated again on the server.

- Every list shows the running count and quantity of accepted scans and allows an accepted scan to be removed before posting.
- Posting buttons are disabled until the mandatory fields for that transaction are complete, and are disabled again while a posting is in progress so it cannot be sent twice.
- Error messages name the rule that refused the action (for example “BR-16”) so that users and support staff can look it up in this document.

8. Data Specifications

M = mandatory, O = optional. Source: where the value comes from — the user, a scan, the device, or the system. Fields shown as "—" are not entered.

8.1 Item / article master

Field	Type	M / O	Source	Validation / rule
Item code	Text 48	M	System	CATEGORY-SUBCAT-SPEC-PO-LINE; unique; never changes
Description	Text 200	M	From PO line	—
Category / sub-category	List	M	From PO line	From the configured category list
Specification / model	Text 60	M	From PO line	Wattage, length, gauge, rating as applicable
Unit of measure	List	M	From PO line	Locked after the first transaction
Tracking	Serialised / Bulk	M	User	Serialised: one label per piece
Pack size	Decimal	M if bulk	User	> 0; quantity carried by one label
PO number and line	Reference	M	System	Must be an Open PO line without an article
Vendor, rate, warranty months	Reference / decimal / integer	M	From PO	Rate ≥ 0; warranty ≥ 0
Expected life (months)	Integer	O	User	≥ 0
Re-order level	Decimal	O	User	≥ 0
Status	Active / Blocked / Obsolete	M	User	Blocked and Obsolete cannot be requisitioned

8.2 Purchase order reference

Field	Type	M / O	Source	Validation / rule
PO number	Text 40	M	User / interface	Unique
Vendor	Reference	M	User / interface	Active vendor
PO date / delivery due	Date	M / O	User / interface	Due date not earlier than PO date
Warranty (months)	Integer	M	User / interface	≥ 0; copied to each article
Line: category, sub-category, specification, description, UoM	Mixed	M	User / interface	As for the article master
Line: ordered quantity / rate	Decimal	M	User / interface	Quantity > 0; rate ≥ 0
Line: received / outstanding	Decimal	—	System	Outstanding = ordered − received
Status	Open / Closed	M	System / user	Closed accepts no receipts

8.3 Store location master

Field	Type	M / O	Source	Validation / rule
Store code / name / type	Text 8 / 120 / list	M	User	Code unique; type Central, Zonal or Sub-store
Zone	List	M	User	Drives data scope
Rack	Text 12	M	User	—
Shelf	Text 12	O	User	May be blank for yard stock

Field	Type	M / O	Source	Validation / rule
Bin	Text 12	M	User	—
Location code	Text 40	—	System	STORE-RACK-SHELF-BIN; unique
Location type	Receiving / Stock / Quarantine / Repair / Scrap	M	User	Only Stock holds issuable material
Active	Yes / No	M	User	Cannot be No while holding stock

8.4 Site master

Field	Type	M / O	Source	Validation / rule
Site ID	Text 32	M	System or survey	Unique; permanent
Site name / description	Text 200	M	User	—
Zone / Division / Ward-UC	List / text	M / M / O	User	Drives data scope
Site type	List	M	User	Road stretch, Junction, Park, Public building, Pole cluster
Latitude / longitude	Decimal (6 dp)	M	User or device	−90…90 / −180…180; not 0,0
Geofence radius (m)	Integer	M	User	≥ 5; default from parameters
Tolerance (m)	Integer	M	Parameter	Default from parameters; per-site override
Assigned supervisor / officer	User reference	O	User	Supervisor controls requisitioning
Status	Planned / Active / Completed / On hold	M	User	Completed and On hold cannot be requisitioned

8.5 QR code payload

Field	Type	M / O	Source	Validation / rule
Prefix	Text	M	System	Fixed value `SLIMS`
Item code	Text 48	M	System	The article's item code
Serial	6 digits	M	System	Sequential within the article, never reused
Check value	6 hex characters	M	System	Derived from item code and serial; a mismatch marks the label as forged
Encoded form	—	—	—	`SLIMS|<item code>|<serial>|<check>`, QR model 2, error correction M
Printed text	—	—	—	Item code, description, PO number, serial and quantity beside the symbol

8.6 Goods receipt

Field	Type	M / O	Source	Validation / rule
GRN number	Text	—	System	GRN/<store>/<FY>/<nnnnn>; sequential
PO number	Reference	M	User	Open PO
PO line, item code, quantity	—	—	From scan	Derived from the label
QR / serial	Scan	M	Scanner	Validated per Section 11
Store	Reference	M	User	In the receiver's scope
Rack / Shelf / Bin	Reference	M at put-away	User / scan	Active Stock location in the same store

Field	Type	M / O	Source	Validation / rule
Receipt date / receiver	Date-time / user	—	System	Server time; signed-in user
Condition	Good / Damaged / Short	M	User	Damaged or Short requires a reason
Challan / invoice number and date	Text / date	O	User	—

8.7 Site requisition

Field	Type	M / O	Source	Validation / rule
Requisition number	Text	—	System	REQ/<zone>/<FY>/<nnnnn>
Site ID	Reference	M	User	Registered, in scope, Active or Planned
Site name	—	—	System	From the site master
Issuing store	Reference	M	User	Active store
Lines: item code / quantity	Reference / decimal	M (≥ 1 line)	User	Active article; quantity > 0
Required date	Date	O	User	Not earlier than today
Priority	Normal / Urgent / Emergency	M	User	Default Normal
Justification / purpose	Text 300	M if Urgent or Emergency	User	—
Requester / date	User / date-time	—	System	—
Approval status	List	—	System	Section 9.2

8.8 Issue to site

Field	Type	M / O	Source	Validation / rule
Issue number / gate pass number	Text	—	System	ISS/<store>/<FY>/<nnnnn>; GP/<store>/<FY>/<nnnnn>
Requisition number	Reference	M	User	Approved or Partly Issued
Site ID / destination	—	—	System	From the requisition
QR / serial	Scan	M	Scanner	Validated per Section 11
Item code / quantity	—	—	From scan	—
Issuing store	—	—	System	From the requisition
Issued by / date-time	—	—	System	—
Handed over to / vehicle	Text	O	User	—
Acknowledged by / at / location	—	—	Device	Recorded by the Lighting Officer

8.9 Installation

Field	Type	M / O	Source	Validation / rule
Installation number	Text	—	System	INST/<site>/<nnnnn>
Client transaction ID	UUID	M	Device	Unique; duplicates ignored
Issue number / Site ID	—	—	System	From the scanned item
Item / QR	Scan	M	Scanner / camera	Issued to this site; not already installed
Installed quantity	Decimal	M	User	1 for serialised items

Field	Type	M / O	Source	Validation / rule
Pole / point identifier	Text 40	O	User	—
Officer	User	—	System	Signed-in user
Installation date-time	Date-time	M	Device	Time of the AFTER capture
BEFORE / AFTER photograph (original)	Image	M	Device camera	Live capture only; within size limit
Latitude / longitude / accuracy / time — per photograph	Decimal / date-time	M	Device	Not editable
Stamped copies and hashes	Image / text	—	System	Section 12.4
Distance to site / evidence status / review status	—	—	System	Sections 12.2, 12.3

8.10 Out-of-order return

Field	Type	M / O	Source	Validation / rule
Return number	Text	—	System	RTN/<FY>/<nnnnn>
Item / QR	Scan	M	Scanner	Installed (for failure / dismantling) or Issued / At Site (unused)
Site ID	—	—	System	From the item
Kind	Failed in service / Dismantled / Unused	M	User	—
Reason / condition	List	M	User	From the reason-code lists
Photographs	Image	O (configurable M)	Device	Live capture
Return date-time / receiving store / location	—	M	System / user	Quarantine by default
Disposition	Usable / Repairable / Not repairable	M after inspection	User	—
Warranty eligible / expires	Yes-No / date	—	System	Installation date + warranty months
Claim reference	Text 60	O	User	—
Disposal approver / date / scrap value	—	M for disposal	System / user	Approving Officer

9. State Models

9.1 Item lifecycle

Every QR-identified item has exactly one status and one location or site at a time. Each status maps to one of the summary statuses of BRD-GSL-001 FR-14, which reports use.

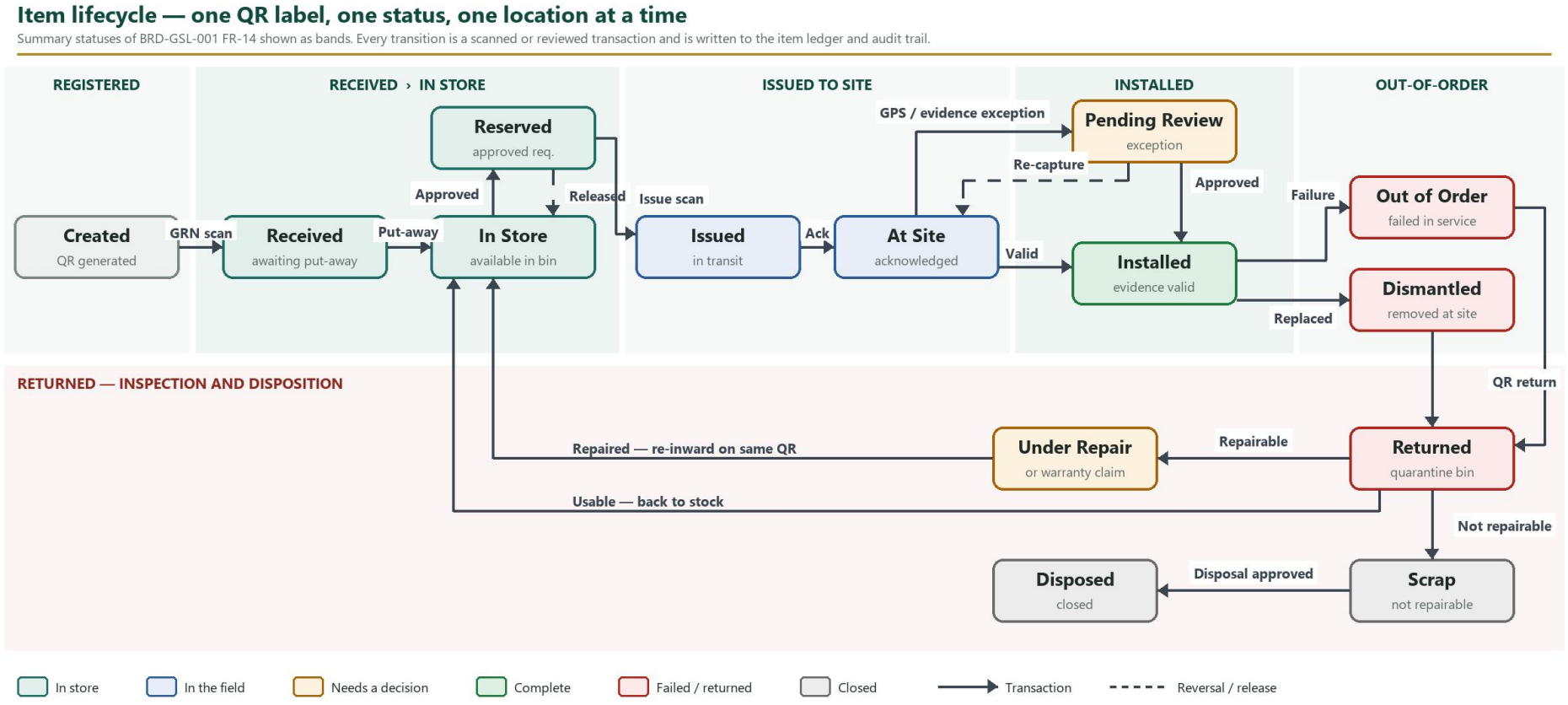


Figure 9-1 — Item lifecycle. Unused returns from site and rejection at receipt are omitted for clarity; see Table 9-1.

Table 9-1 — Item statuses and permitted transitions

Status	Meaning	Summary status (GFR-14)	Entered by	Permitted next status
Created	Label generated against a PO line; not yet received	Registered	Article creation	Received, Rejected, Cancelled
Received	Received at the store, in the Receiving location	Received	GRN scan	In Store, Rejected
In Store	At a Stock bin, available	In Store	Put-away, re-inward, usable return	Reserved, Issued to Site, Out of Order*
Reserved	Committed to an approved requisition	In Store	Requisition approval	Issued to Site, In Store (release)
Issued to Site	Custody passed to the site; in transit	Issued to Site	Issue posting	At Site, In Store (unused return)
At Site	Acknowledged by the Lighting Officer	Issued to Site	Acknowledgement	Installed, Pending Review, In Store (unused return)
Pending Review	Installation submitted; evidence not valid	Issued to Site	Validation (exception)	Installed, At Site (re-capture / rejected)
Installed	Installation complete with valid or approved evidence	Installed	Validation or approved exception	Out of Order, Dismantled
Out of Order	Failed in service at the site	Out-of-Order	Out-of-order report	Returned
Dismantled	Removed at the site during replacement or shifting	Out-of-Order	Out-of-order report	Returned
Returned	Scanned back into Quarantine; awaiting inspection	Returned	Return receipt	In Store, Under Repair, Warranty Claim, Scrap
Under Repair / Warranty Claim	With a repairer or the vendor	Returned	Classification	In Store (re-inward), Scrap
Scrap	Condemned; in the Scrap location	Returned	Classification	Disposed
Disposed	Written out of stock. Terminal.	Closed	Disposal approval	—
Rejected / Cancelled	Rejected at receipt, or label cancelled	Registered	Receipt rejection	Received (after rectification), —

** An item in store may be reported damaged and moved to Quarantine through an approved adjustment.*

9.2 Requisition status

Status	Meaning	Permitted next status
Draft	Being prepared by the requester	Submitted, Cancelled
Submitted	Awaiting the Approving Officer	Approved, Rejected, Escalated, Cancelled
Escalated	Above the officer's limit; with the next level	Approved, Rejected
Approved	Stock reserved; Issue Slip generated	Partly Issued, Issued, Lapsed, Cancelled
Rejected	Returned with a reason	Draft (revised as a new version)
Partly Issued	Some lines or quantity issued	Issued, Lapsed
Issued	Every approved line satisfied. Terminal.	—
Lapsed	Reservation validity expired before issue	Submitted (re-submitted)
Cancelled	Cancelled before issue. Terminal.	—

9.3 Installation evidence and review status

Two separate statuses describe an installation. The **evidence status** is the system's verdict on the evidence; the **review status** is the human decision on anything that is not Valid. BRD-GSL-001 §9 lists Valid, Location Exception, Missing Evidence and Pending Review; the last is the review state.

Status type	Value	Meaning	Consequence / next
Evidence status	Valid	Both photographs present and within geofence + tolerance	Installation Complete; item Installed
Evidence status	Location Exception	A photograph lies outside geofence + tolerance	Review status Pending Review
Evidence status	Missing Evidence	A photograph is missing or fails its integrity check after synchronisation	Review status Pending Review; officer asked to re-capture
Review status	Pending Review	In the Project / Monitoring Officer's queue	Approved, Re-capture Required, Rejected
Review status	Approved	Deviation accepted with a written justification	Installation Complete; item Installed
Review status	Re-capture Required	Officer must capture new evidence	A new installation transaction supersedes this one
Review status	Rejected	Installation void	Item remains At Site pending action

9.4 Return disposition

Stage	Meaning	Actor	Next
Reported	Out of order or dismantled at site	Officer	Received
Received	In the Quarantine location	Store / Warehouse Officer	Usable, Repairable, Not repairable
Usable	Fit for use without repair	Store / Warehouse Officer	Closed — back in stock
Repairable	Under repair or warranty claim	Store / Warehouse Officer; Procurement / Admin	Closed — re-inwarded; or Not repairable
Not repairable	In the Scrap location	Store / Warehouse Officer	Closed — disposed after approval

10. Business Rules

BR-01 to BR-10 are the key business rules of BRD-GSL-001 §15, in order. BR-11 onwards come from the SLIMS BRD and BRD-GSL-001 body text. Every rule is enforced by the service interface; the user interface only anticipates it.

ID	Rule	Enforced at	On violation	Source
BR-01	No PO / reference → no standard receipt.	Goods receipt	Blocked — VAL-INV-001	G Rule 1
BR-02	No valid item / QR → no controlled stock movement.	Every scan	Blocked — VAL-INV-003 / 004	G Rule 2; S BR-001
BR-03	No approved requisition → no issue to site.	Issue	Blocked — VAL-INV-020	G Rule 3; S BR-006
BR-04	An issue must identify a registered Site ID.	Requisition, issue	Blocked — VAL-REQ-001	G Rule 4; S BR-007
BR-05	Installation requires before and after evidence.	Installation	Blocked — VAL-FLD-003	G Rule 5; S BR-013
BR-06	A GPS mismatch creates an exception; it never completes an installation.	Validation	Exception created	G Rule 6; S BR-017
BR-07	An out-of-order return requires a QR scan and a condition / reason.	Returns	Blocked — VAL-INV-030	G Rule 7
BR-08	Every stock movement updates the item ledger.	All stock postings	Posting fails if the ledger entry fails	G Rule 8
BR-09	Every exception or override requires authorised review and an audit trail.	Exceptions, over-receipt, adjustments, disposal, amendments	Blocked without approver and reason	G Rule 9
BR-10	A site must exist in the system before material is issued to it.	Requisition, issue	Blocked — VAL-REQ-001	G Rule 10
BR-11	A QR identifier is unique system-wide and never reused.	Label generation, every scan	Blocked — VAL-INV-006	S BR-002
BR-12	The item code carries its PO reference for the item's whole life.	Article creation	Article cannot be saved without a PO line	S BR-003, BR-004
BR-13	Receipt may not exceed the PO line balance without an approved over-receipt exception.	Goods receipt	Blocked — VAL-INV-007	G §5.2; S BR-014
BR-14	Stock is always at a location, and available only after put-away to a Stock bin.	Receipt, put-away	Blocked — VAL-INV-008 / 009	G §7; S BR-005
BR-15	Approved quantity is reserved and cannot be issued against another requisition.	Approval, issue	Blocked — VAL-INV-024	S BR-009
BR-16	Issued quantity may never exceed the approved quantity.	Issue	Blocked — VAL-INV-022	S BR-010
BR-17	An item issued to a site is unavailable for another issue until an authorised transaction changes its status.	Issue	Blocked — VAL-INV-023	G §10; S BR-011
BR-18	A Lighting Supervisor may requisition only for sites in their scope.	Requisition	Blocked — VAL-REQ-002	G §4; S BR-008
BR-19	Photographs are captured live; gallery selection and upload are not possible.	Installation	Not offered by the app	S BR-015
BR-20	Coordinates and times come from the device at capture and cannot be edited.	Installation	Blocked — VAL-FLD-004	S BR-016
BR-21	The original photograph is preserved; the stamped copy is generated by the system.	Installation	—	G §9
BR-22	Returned failed items enter Quarantine, never issuable stock directly.	Returns	Location forced to Quarantine	S BR-018
BR-23	Warranty eligibility is computed by the system, not chosen by the user.	Returns	Blocked — VAL-INV-032	S BR-019

ID	Rule	Enforced at	On violation	Source
BR-24	Disposal requires recorded approval.	Disposal	Blocked — VAL-INV-033	S BR-020
BR-25	No transaction may make stock negative.	All stock postings	Blocked — VAL-INV-010	S BR-021
BR-26	No posted transaction is deleted; corrections are reversals or approved amendments.	All	Delete not offered	G §12; S BR-022
BR-27	No user approves or reviews a transaction they created.	Approvals, reviews	Blocked — VAL-REQ-005, VAL-EXC-002, VAL-SEC-003	S BR-026
BR-28	Stock adjustments require a reason and an approver.	Adjustments	Blocked without both	G §10
BR-29	Offline transactions keep their original capture time and coordinates.	Synchronisation	—	S BR-027
BR-30	A transaction ID is accepted once.	All postings	Duplicate ignored — VAL-FLD-007	G §14
BR-31	Changing a site's coordinates after material has been issued to it requires approval.	Site master	Held for approval	S BR-025
BR-32	A store or location holding stock cannot be deactivated.	Store master	Blocked — VAL-MDM-003	S BR-024
BR-33	Removed items must reach a store within the configured return period.	Returns	Flagged and escalated	S BR-029

11. Validation and Error Messages

The service interface returns each refusal as a message and its ID; the user interface shows both. Text in braces is filled in from the transaction.

ID	Condition	Message shown to the user	Behaviour
VAL-MDM-001	PO line already has an article	Article {code} already exists for this PO line.	Block
VAL-MDM-002	Label count above limit	This would generate {n} labels; the limit is {max}. Use a larger pack size.	Block
VAL-MDM-003	Deactivating a location with stock	{location} holds stock and cannot be deactivated.	Block
VAL-INV-001	No PO selected	Select an open purchase order. Material cannot be received without a PO reference.	Block
VAL-INV-002	PO closed	PO {po} is closed and accepts no further receipts.	Block
VAL-INV-003	Check value fails	This is not a valid SLIMS label. Do not receive it; report the label.	Block
VAL-INV-004	Label not registered	Label {item code} #{serial} is not registered in the system.	Block
VAL-INV-005	Label from another PO	This item belongs to PO {po}, not the PO being received.	Block
VAL-INV-006	Label already received	Already {status} — a label can only be received once.	Block
VAL-INV-007	Over PO balance	Receipt would exceed the PO line balance ({received} + {qty} > {ordered}). Request an over-receipt exception to continue.	Block
VAL-INV-008	No bin chosen	Choose the Store ▸ Rack ▸ Shelf ▸ Bin location.	Block
VAL-INV-009	Bin is not a Stock location	Material can only be put away to an active Stock location.	Block
VAL-INV-010	Negative stock	This would take stock at {location} below zero.	Block
VAL-INV-020	Requisition not issuable	Requisition {no} is {status}; material can only be issued against an approved requisition.	Block
VAL-INV-021	Item not on requisition	{item code} is not on requisition {no}.	Block
VAL-INV-022	Above approved quantity	Excess — {approved} {uom} approved, {issued} already issued.	Block
VAL-INV-023	Item not available	Item is {status} ({site or location}) and cannot be issued.	Block
VAL-INV-024	Reserved elsewhere	This item is reserved for requisition {other}.	Block
VAL-INV-025	Item at another store	Item is at store {store}, not the issuing store.	Block
VAL-INV-026	Scanned twice in one transaction	Already scanned in this transaction.	Ignore
VAL-INV-030	Return without reason	Select the failure reason and condition before submitting.	Block
VAL-INV-031	Item not installed	Item is {status}; only an installed item can be reported out of order.	Block
VAL-INV-032	Warranty expired	Warranty expired on {date}. Route the item as Repairable or Not repairable.	Block
VAL-INV-033	Disposal before condemnation	Only items classified Not repairable can be disposed.	Block
VAL-REQ-001	No registered site	Select a site from the site master. Free-text sites are not accepted.	Block
VAL-REQ-002	Site out of scope	Site {site} is outside your authorised scope.	Block
VAL-REQ-003	Site not active	Site {site} is {status}; material cannot be requisitioned for it.	Block
VAL-REQ-004	Shortfall	Available {qty} {uom}; {short} will be recorded as a purchase indent.	Warn
VAL-REQ-005	Self-approval	You cannot approve a requisition you raised.	Block
VAL-REQ-006	Reject without reason	A reason is required to reject.	Block
VAL-REQ-007	Above approval limit	Value {value} exceeds your limit; the requisition has been escalated to {role}.	Info
VAL-FLD-001	Item not issued to a site	This item has not been issued to any site.	Block

ID	Condition	Message shown to the user	Behaviour
VAL-FLD-002	Site not assigned to officer	This item was issued to site {site}, which is not assigned to you.	Block
VAL-FLD-003	Photograph missing	Both BEFORE and AFTER photographs are required.	Block
VAL-FLD-004	No coordinates	The photograph carries no location. Enable location and capture it again.	Block
VAL-FLD-005	Poor accuracy	GPS accuracy ±{acc} m is poorer than the ±{min} m standard. Wait for a better fix.	Warn; Block beyond 3×
VAL-FLD-006	Untrusted location or time	The device location or clock cannot be trusted. The submission was refused and logged.	Block
VAL-FLD-007	Duplicate submission	This installation was already received (transaction {id}). No duplicate was created.	Info
VAL-EXC-001	Short justification	A written justification of at least {n} characters is required to approve an exception.	Block
VAL-EXC-002	Reviewing own transaction	You cannot review an exception on your own installation.	Block
VAL-SEC-001	Out of scope	{record} is outside your scope.	Block
VAL-SEC-002	No permission	Your role ({role}) cannot perform this action.	Block
VAL-SEC-003	Approving own transaction	You cannot approve a transaction you raised.	Block

12. GPS, Evidence and Geofence Specification

12.1 Capture

- Both photographs are taken through the application's camera; the device gallery is never offered.
- At the instant of each capture the app records latitude, longitude, horizontal accuracy and device time, and reads the server-synchronised time offset.
- The app will not capture while accuracy is worse than three times the configured minimum, and warns between the minimum and three times the minimum.
- The site ID, item code, officer and client transaction ID are bound to the capture; none can be changed afterwards.
- Captured originals are stored on the device encrypted until synchronised, and deleted from the device once the server confirms receipt.

12.2 Distance calculation

Distance between the site point and a photograph is the great-circle (Haversine) distance: $d = 2R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)})$, where $a = \sin^2(\Delta\phi/2) + \cos\phi_1 \cdot \cos\phi_2 \cdot \sin^2(\Delta\lambda/2)$, ϕ is latitude and λ longitude in radians, and $R = 6{,}371{,}000$ m. The result is stored to one decimal place on the installation.

Worked example from acceptance testing: for a road-stretch site with a 250 m geofence and 20 m tolerance, evidence 16 m from the site point is Valid; evidence 2,922 m away is a Location Exception.

12.3 Status determination

1. If either photograph is missing, or its hash does not match the value the device sent, set evidence status **Missing Evidence** and go to step 6.
2. If the device reported a mock location provider, or the capture time differs from server time by more than the permitted drift (online captures), refuse the submission and log it (VAL-FLD-006). No installation is created.
3. Compute d(before) and d(after), the distances from the site point to each photograph (Section 12.2). Let d = the larger.
4. Allowed = site geofence radius + tolerance.
5. If d ≤ Allowed, set evidence status **Valid**, mark the installation **Complete**, set the item to **Installed**, and stop.
6. Otherwise (or from step 1) set evidence status **Location Exception** (or **Missing Evidence**), review status **Pending Review**, item status **Pending Review**, notify the Project / Monitoring Officer, and place the installation in the exception queue.

12.4 Stamping and integrity

- The original photograph is stored exactly as captured. The device computes its SHA-256 hash at capture; the server recomputes it on receipt and stores both.
- The server generates the stamped copy from the recorded metadata — never from typed input — with a band at the foot of the image showing local date and time, latitude and longitude to six decimals, accuracy, Site ID, item code, officer and installation number.
- Screens and reports show the stamped copy by default, with the original one click away. Neither can be edited or replaced after submission.
- If a stored original no longer matches its hash, the evidence status becomes Missing Evidence and an exception is raised.

12.5 Configurable parameters

Parameter	Default	Set by	Note
Default geofence radius	50 m	System Administrator	Per-site override; larger for road stretches, smaller for a single pole
Tolerance	20 m	System Administrator	Allowance for GPS drift in built-up areas
Minimum GPS accuracy	20 m	System Administrator	Warning below this; capture refused beyond 3×
Maximum clock drift	5 minutes	System Administrator	Online captures only; offline captures use the synchronised offset
Offline retention	7 days	System Administrator	Unsynchronised transactions older than this are flagged
Maximum photograph size	8 MB	System Administrator	After compression
Minimum justification length	10 characters	System Administrator	For approving an exception
Return period after removal	7 days	System Administrator	BR-33
Acknowledgement overdue	48 hours	System Administrator	Issue not acknowledged at site
Reservation validity	5 working days	System Administrator	Unissued reservations released

13. Stock Ledger and Reconciliation

13.1 Movement types

Movement	Trigger	On hand	Available	Location
Receipt	GRN posted	+ qty (Receiving)	No change	— → Receiving (or bin)
Put-away	Item scanned to bin	No change	+ qty	Receiving → Stock bin
Bin move	Internal move	No change	No change	Bin → bin
Reservation	Requisition approved	No change	− qty	—
Release	Cancel, lapse or surplus on issue	No change	+ qty	—
Issue	Issue posted	− qty	— (reserved quantity consumed)	Stock bin → Site
Unused return	Unused material back to stock	+ qty	+ qty	Site → Stock bin
Failed return	Return received	+ qty (Quarantine)	No change	Site → Quarantine
Usable / re-inward	Classified Usable, or repaired	No change	+ qty	Quarantine / Repair → Stock bin
Transfer out / in	Despatch / receipt legs	− / + qty	− / + qty	Store A → in transit → Store B
Adjustment	Approved adjustment	± qty	± qty	—
Disposal	Disposal approved	− qty (Scrap)	No change	Scrap → written off

13.2 Reconciliation

Opening + Receipts + Returns − Issues − Approved Adjustments = Closing

- **Opening** and **Closing** are on-hand quantities at the start and end of the period, including Receiving, Quarantine, Repair and Scrap locations.
- **Receipts** are GRN postings net of reversals. **Returns** are unused and failed returns received into any location of the store. **Issues** are issue postings. **Approved Adjustments** are signed, and include disposals and the net of transfers.
- The reconciliation is produced per item code and per store, and is reported with any line that does not balance highlighted. An unbalanced line is itself an exception.

14. Notifications and Alerts

In-app notifications are always sent; e-mail and SMS where the recipient has them configured. Escalation periods are parameters.

ID	Event	Recipients	Channel	Escalation
NTF-01	Requisition submitted	Approving Officer	In-app, e-mail	Escalate to next level after 24 h
NTF-02	Requisition approved	Lighting Supervisor; issuing Store / Warehouse Officer	In-app, SMS	—
NTF-03	Requisition rejected	Lighting Supervisor	In-app, SMS	—
NTF-04	Reservation lapsing in 1 day / lapsed	Lighting Supervisor; Store / Warehouse Officer	In-app	—
NTF-05	Issue completed (gate pass)	Lighting Supervisor; Lighting Officer of the site	In-app, SMS	—
NTF-06	Issue not acknowledged	Lighting Officer; Lighting Supervisor	In-app, SMS	After acknowledgement period
NTF-07	Installation pending — issued but not installed	Lighting Officer; Lighting Supervisor	In-app	Monitoring Officer after 7 days
NTF-08	GPS exception raised	Project / Monitoring Officer; copy Lighting Supervisor	In-app, e-mail	Escalate after 3 days unresolved
NTF-09	Missing evidence / re-capture required	Lighting Officer	In-app, SMS	Supervisor after 2 days
NTF-10	Field verification requested	Lighting Supervisor	In-app, SMS	—
NTF-11	Out-of-order return reported	Store / Warehouse Officer of the receiving store	In-app	—
NTF-12	Return overdue	Lighting Supervisor; Project / Monitoring Officer	In-app, e-mail	Weekly until resolved
NTF-13	Over-receipt exception / adjustment / disposal awaiting approval	Approving Officer	In-app, e-mail	Escalate after 24 h
NTF-14	Stock below re-order level	Store / Warehouse Officer; Procurement / Admin	In-app, e-mail	Daily digest
NTF-15	Repeated GPS mismatch (≥ 3 in 30 days by one officer or at one site)	Project / Monitoring Officer	In-app, e-mail	—
NTF-16	Unusual return rate (article or vendor above threshold)	Procurement / Admin; Management	E-mail	Monthly
NTF-17	Unresolved site installations (site with issued material and no installation for 14 days)	Project / Monitoring Officer	In-app	Weekly

15. Reports and Dashboards

15.1 Report catalogue

Every report accepts the common filters of FR-RPT-003, is limited to the viewer's scope and exports to Excel and PDF.

ID	Report	Content	Main filters	Drills to
RPT-01	PO receipt register	GRNs with PO, vendor, challan, quantity, value, bin	Date, PO, vendor, store	GRN → items
RPT-02	Discrepancy & over-receipt register	Short, excess, rejected and over-received quantities; resolution; age	PO, vendor, status	Note → GRN
RPT-03	Stock position	On hand, reserved, available, quarantine, in transit by bin	Store, location, item, category	Bin → items → history
RPT-04	Stock reconciliation	Opening, receipts, returns, issues, adjustments, closing; imbalances	Store, item, period	Line → ledger entries
RPT-05	Requisition status & turnaround	Requested, approved, issued, balance; hours to decision and to issue	Site, supervisor, status, priority	Requisition → issue
RPT-06	Issue register & gate passes	Issues with site, items, collector, vehicle, acknowledgement	Store, site, date	Issue → items
RPT-07	Site installation report	Installations by site, pole, item, officer, date, result	Area, site, officer, date	Site → installation → evidence
RPT-08	GPS validation report	Distance, allowed limit, accuracy, evidence status for every installation	Area, officer, status	Installation → evidence
RPT-09	Evidence register	Every photograph with coordinates, accuracy, capture and upload time, hash	Site, officer, date	Photograph (original / stamped)
RPT-10	Exception register & ageing	Exceptions by type, age, reviewer, decision, justification	Type, area, status, age	Exception → evidence
RPT-11	Out-of-order & return report	Returns by reason, condition, disposition, days in service	Area, site, article, vendor	Return → item history
RPT-12	Warranty claim register	Claims raised, vendor, outcome, value recovered	Vendor, PO, status	Claim → item
RPT-13	Scrap & disposal register	Condemned items, approver, scrap value	Store, period	Disposal → item history
RPT-14	Site-wise progress	Sanctioned, issued, installed, pending, failed per site	Area, site, status	Site → requisitions → installations
RPT-15	Item-wise consumption	Issued and installed quantity and value by article	Article, category, area, period	Article → sites
RPT-16	PO utilisation	Ordered, received, in stock, issued, installed, failed, balance by PO line	PO, vendor	Line → items
RPT-17	Officer performance	Installations, first-time valid rate, exceptions, returns by officer	Officer, area, period	Officer → installations
RPT-18	Stock ageing	Stock by age band per bin and PO	Store, article, band	Bin → items
RPT-19	Item traceability	PO → GRN → locations → requisition → issue → installation → return → status	QR, item code	Every linked record
RPT-20	Audit trail extract	Every action with user, role, time, address, before / after values	User, action, entity, date	Entry → record

15.2 Dashboard

Widget	Measure	Drills to
Stock received	Quantity and value received in the period	RPT-01

Widget	Measure	Drills to
Available stock	Available quantity and value by store	RPT-03
Issued to site	Quantity issued and not yet installed	RPT-06, RPT-14
Installed	Installations complete in the period	RPT-07
Pending installation	Issued material not installed, by age	RPT-14
GPS-validated installations	First-time Valid as a share of all installations	RPT-08
GPS exceptions	Open exceptions by type and age	RPT-10
Out-of-order returns	Reported, received, overdue	RPT-11
Site-wise progress	Progress by area and site	RPT-14
Item-wise consumption	Consumption by article	RPT-15
PO utilisation	Received and installed against ordered	RPT-16
Audit trail	Latest actions in scope	RPT-20

15.3 Drill-down path

Summary → Administrative area → Site → Requisition / Issue → Item → Evidence / validation record

16. Access Control Matrix

Function	Proc. / Admin	Store / WH Officer	Lighting Supervisor	Approving Officer	Lighting Officer	Project / Monitoring	Mgmt / Dashboard	System Admin
PO reference & vendors	C R U	R	R	R	—	R	R	R
Article creation & QR series	C R	C R	—	R	—	R	R	R
QR label print / re-print	C	C	—	—	—	—	—	R
Store & bin master	R	C R U [1]	—	R	—	R	R	C R U
Site master	R	R	C R U [2]	R	R	R	R	C R U
Goods receipt & put-away	R	C R U [1]	—	R	—	R	R	R
Over-receipt exception	R	C	—	A	—	R	R	—
Discrepancy note	U [3]	C	—	R	—	R	R	R
Transfer, bin move	—	C [1]	—	R	—	R	R	R
Stock adjustment & verification	—	C [1]	—	A	—	R	R	R
Requisition create / submit	—	R	C R U [2]	R	R	R	R	R
Requisition approve / reject	—	R	R	A	—	R	R	R
Issue to site	—	C [1]	R	R	R	R	R	R
Acknowledge at site	—	R	R	—	C [2]	R	R	R
Installation & evidence	—	—	R	—	C [2]	R	R	R
Exception review	—	—	U [4]	—	R	A	R	R
Report out-of-order	—	R	R	—	C [2]	R	R	R
Receive & classify return	—	C [1]	R	R	R	R	R	R
Warranty claim	C U	R	—	R	—	R	R	R
Repair re-inward	—	C [1]	—	R	—	R	R	R
Disposal approval	—	C	—	A	—	R	R	R
Amendment of a completed record	—	C	C	A [5]	—	A [5]	R	R
Reports & dashboards	R	R	R	R	R	R	R	R
Audit trail	—	R [1]	—	R	—	R	R	R
Users, roles, approval matrix, parameters	—	—	—	—	—	—	—	C R U

C = create / perform, R = read, U = update, A = approve, — = no access. [1] assigned stores only. [2] assigned sites only. [3] Procurement / Admin records the resolution agreed with the vendor. [4] field-verification findings only, when requested by the reviewer. [5] stock records by the Approving Officer; installation records by the Project / Monitoring Officer. All access is further limited to the user's data scope (Section 4.2).

17. Audit Trail

17.1 Recorded for every action

Field	Content
Audit ID	Sequential, never reused
Transaction ID	The business transaction the action belongs to (GRN, requisition, issue, installation, return…)
Entity type and ID	Item, article, PO, requisition, issue, installation, return, site, store, user, parameter
Action	For example received, reserved, issued, installed, exception-approved, disposed, login-failed
User ID, name and role	The signed-in user at the time
Server time	UTC; displayed in local time
Device capture time	For field transactions, alongside server time
Client address and device	Taken from the trusted reverse proxy; forged headers are discarded
Before and after values	Changed fields only
Reason / justification	Mandatory for overrides, rejections, adjustments, disposals and amendments
Evidence reference	Photograph IDs where evidence is involved

17.2 Protection and retention

- The application has no function to update or delete an audit entry; database permissions for the application account allow insert and read only.
- Transactions are retained for ten years; photographs for at least five years, or longer if the records-retention policy requires. Evidence cannot be purged before its retention date.
- Failed sign-ins, refused scans and refused actions are audited as well as successful ones, so that attempts are visible.

18. Interfaces

18.1 Service interface

All screens use one service interface; access control and business rules are applied there. The functional grouping below is taken from the reference build. Every refusal is returned as { error, step }, where step is the rule or validation ID.

Function	Operation	Roles
Sign in, refresh, sign out	POST /api/auth/login · /refresh · /logout	All
Articles and QR series; label sheet	GET, POST /api/articles · POST /api/articles/{id}/labels	Procurement / Admin, Store
Item enquiry and trace	GET /api/items/{id or QR payload}	All (scope)
Purchase orders	GET, POST, PUT /api/purchase-orders	Procurement / Admin
Stores and bins	GET, POST, PUT /api/stores · /api/stores/{id}/locations	Store, System Administrator
Sites; bulk upload	GET, POST, PUT /api/sites · POST /api/sites/bulk	Supervisor, System Administrator
Goods receipt: validate scan, post, reverse	POST /api/grn/scan · POST /api/grn · POST /api/grn/{id}/reverse	Store
Discrepancy notes	GET, POST /api/grn/discrepancies · POST …/{id}/resolve	Store, Procurement / Admin
Requisitions: create, submit, decide, cancel	POST /api/requisitions · /{id}/submit · /{id}/decide · /{id}/cancel	Supervisor, Approving Officer
Issue: validate scan, post, acknowledge, work list	POST /api/issues/scan · POST /api/issues · POST /api/issues/{id}/acknowledge · GET /api/issues/field	Store, Lighting Officer
Installation submit (multipart); exception decision	POST /api/installations · POST /api/installations/{id}/verify	Lighting Officer, Monitoring Officer
Evidence file	GET /api/files/{id}	All (scope)
Returns: report, receive, classify, re-inward, dispose	POST /api/returns/report · /receive · /{id}/classify · /{id}/reinward · /{id}/dispose	Officer, Store, Approving Officer
Stock position; dashboard	GET /api/stock · GET /api/dashboard	All (scope)
Reports	GET /api/reports · GET /api/reports/{key}	All (scope)
Audit trail	GET /api/audit	Monitoring, Management, System Administrator
Users, roles, passwords; parameters	GET, PUT /api/users · POST /api/users/{id}/password · GET, PUT /api/config	System Administrator

Functions required by this FRD that have no endpoint in the reference build yet: put-away and bin moves, over-receipt exception, transfers, adjustments and physical verification, usable-return classification, amendment requests, notifications, and client transaction IDs on every posting (Annexure A).

18.2 External interfaces

ID	System	Direction	Data	Mode	Pri.
INT-01	Procurement / PO system	Inbound	Approved PO header, lines, rates and warranty terms	API near real time; file upload as fallback	M
INT-02	Procurement / PO system	Outbound	Purchase indents, discrepancy notes, warranty claims	API	S
INT-03	Finance / accounting	Outbound	GRN values, consumption, disposals, period stock valuation	Scheduled file	S
INT-04	SMS and e-mail gateway	Outbound	Notifications (Section 14)	API	M

ID	System	Direction	Data	Mode	Pri.
INT-05	Map service	Inbound	Base map for site and installation maps	API	S
INT-06	Label printer	Outbound	QR label print jobs	Local driver / print API	M
INT-07	Directory / single sign-on	Inbound	Authentication and organisational identity	SAML or OIDC	C

19. Non-Functional Requirements with Functional Impact

Only the non-functional requirements that change how a function behaves are listed; the full set is in the SLIMS BRD Section 17.

ID	Area	Requirement	Affects	Source
NFR-01	Scan response	A scan is validated and the result shown within 2 seconds	Scanning must keep pace with handling material	G §14
NFR-02	Screen response	95 % of screens respond within 3 seconds	—	S NFR-02
NFR-03	Report response	A year of data returns within 30 seconds; longer reports are queued	—	G §14
NFR-04	Responsive interface	Store and office screens work on desktop; field screens on Android 10+ phones, outdoors and with gloves	—	G §14
NFR-05	Offline capture	Field capture and scanning work offline for the retention window with no data loss	FR-FLD-012	G §14
NFR-06	Controlled synchronisation	Synchronisation is idempotent and preserves evidence integrity (hash check)	FR-FLD-013, FR-SEC-007	G §14
NFR-07	Time synchronisation	Servers synchronise to a trusted time source; device drift is measured on every sync	FR-FLD-011	G §14
NFR-08	Transport security	TLS 1.2 or higher for all traffic; refresh cookie never sent without TLS	—	G §14
NFR-09	Availability	99.5 % monthly in operational hours	—	S NFR-06
NFR-10	Backup and recovery	RPO ≤ 15 minutes, RTO ≤ 4 hours; database and evidence store backed up together	—	G §14
NFR-11	Scalability	≥ 500,000 labels, ≥ 50,000 sites, ≥ 5 million transactions a year	—	S NFR-04
NFR-12	Configurability	Geofence, tolerance and thresholds changeable without code changes	FR-MDM-020	G §14
NFR-13	Language	English and Urdu labels and printed documents	—	S NFR-10

20. Traceability Matrix

20.1 BRD-GSL-001 functional requirements

G FR	Title	FRD requirements	Use cases	Test cases
FR-01	PO-based receiving	FR-MDM-009; FR-INV-001–002	UC-02	TC-01, TC-03
FR-02	QR receiving	FR-INV-002	UC-02	TC-01, TC-03
FR-03	Put-away	FR-MDM-012; FR-INV-011	UC-03	TC-01
FR-04	Digital requisition	FR-REQ-001	UC-05	—
FR-05	Approval workflow	FR-REQ-005, 009, 013	UC-05, UC-06	TC-08, TC-10, TC-11
FR-06	Issue to site	FR-INV-023	UC-07	TC-13
FR-07	No unauthorised issue	FR-INV-022–023; FR-REQ-007	UC-07	TC-12, TC-13
FR-08	Installation transaction	FR-FLD-002–003	UC-09	TC-18
FR-09	Evidence	FR-FLD-004–006	UC-09, UC-10	TC-16
FR-10	Location capture	FR-FLD-008	UC-09	—
FR-11	GPS validation	FR-EXC-001	UC-10	TC-18
FR-12	Exception handling	FR-INV-006; FR-EXC-004–005	UC-02, UC-11	—
FR-13	Out-of-order return	FR-INV-030–032	UC-12, UC-13	TC-23, TC-24
FR-14	Status history	FR-MDM-007; FR-INV-014; FR-RPT-006	UC-15	TC-28
FR-15	Reports	FR-RPT-001–002	—	—

20.2 BRD-GSL-001 key business rules

G rule	Rule	FRD rule	FRD requirements	Test cases
1	No PO / reference → no standard receipt.	BR-01	FR-INV-001, 003	TC-01, TC-02
2	No valid item / QR → no controlled stock movement.	BR-02	FR-MDM-006; FR-INV-002	TC-01, TC-03, TC-04
3	No approved requisition → no issue to site.	BR-03	FR-INV-022	TC-12
4	Issue must identify a registered Site ID.	BR-04	FR-INV-025; FR-REQ-002	TC-07, TC-15
5	Installation requires before / after evidence.	BR-05	FR-FLD-004–006	TC-16
6	GPS mismatch creates an exception.	BR-06	FR-EXC-003	TC-19
7	Out-of-order return requires QR scan and condition / reason.	BR-07	FR-INV-030–031	TC-23
8	Every stock movement updates the item ledger.	BR-08	FR-INV-013	TC-01, TC-15, TC-29
9	Every exception / override requires authorised review and audit trail.	BR-09	FR-INV-004, 020, 037; FR-REQ-011; FR-EXC-006, 008–009	TC-05, TC-09, TC-20, TC-21, TC-26
10	Site locations must exist in the system before site issuance.	BR-10	FR-MDM-015; FR-INV-025; FR-REQ-002	TC-07, TC-15

21. Functional Acceptance Test Cases

These cases prove the acceptance criteria of BRD-GSL-001 §16: that an article can be followed from PO to final status, and that unauthorised, out-of-scope, mismatched and duplicate actions are refused.

ID	Scenario	Steps	Expected result	Requirements	Rules
TC-01	Receive a PO-referenced article by QR into a bin	Select store and PO; scan 5 labels; choose a Stock bin; post	GRN numbered; items In Store at the bin; PO outstanding reduced by 5; ledger shows 5 receipts	FR-INV-002, 007, 011	BR-01, BR-08
TC-02	Receive without a PO	Open receipt and scan without selecting a PO	Refused — VAL-INV-001	FR-INV-003	BR-01
TC-03	Scan a forged label	Scan a payload with a wrong check value	Refused — VAL-INV-003; attempt audited	FR-INV-002; FR-MDM-006	BR-02
TC-04	Receive the same label twice	Receive a label, then scan it on a new receipt	Refused — VAL-INV-006	FR-MDM-006	BR-11
TC-05	Receive beyond PO balance	Scan one more label than outstanding; then request and approve an over-receipt exception	First refused — VAL-INV-007; after approval the item is received and the exception is linked to the GRN	FR-INV-004	BR-13, BR-09
TC-06	Stock not yet put away is not available	Receive to Receiving; open a requisition for the article	Receiving quantity is not shown as available	FR-INV-010	BR-14
TC-07	Requisition for an unregistered or out-of-scope site	Try to type a site name; try to select another supervisor's site	No free-text field; out-of-scope site not selectable — VAL-REQ-002	FR-REQ-002	BR-04, BR-18
TC-08	Submit a requisition with a shortfall	Request more than available; submit	Warning VAL-REQ-004; requisition Submitted; indent recorded; Approving Officer notified	FR-REQ-005–006; FR-NTF-001	—
TC-09	Approve own requisition	As a user with both rights, raise and approve	Refused — VAL-REQ-005	FR-REQ-011	BR-27
TC-10	Approve in part	Approve 3 of 5 requested	3 reserved; Issue Slip numbered; supervisor notified	FR-REQ-009, 012	BR-15
TC-11	Reject a requisition	Reject without reason; then with reason	First refused — VAL-REQ-006; then returned to supervisor with the reason	FR-REQ-009, 013	—
TC-12	Issue before approval	Try to issue against a Submitted requisition	Refused — VAL-INV-020	FR-INV-022	BR-03
TC-13	Issue above approved quantity	Scan one more label than approved	Refused — VAL-INV-022	FR-INV-023	BR-16
TC-14	Duplicate QR movement	Scan a label already issued to another site	Refused — VAL-INV-023 with its current site shown	FR-INV-024	BR-17
TC-15	Post an issue	Scan approved items; post	Issue Note and Gate Pass numbered; items at the site; bins relieved; ledger updated	FR-INV-013, 025–026	BR-08
TC-16	Installation without the AFTER photograph	Capture BEFORE only; submit	Not submittable — VAL-FLD-003	FR-FLD-004, 006	BR-05
TC-17	Gallery photograph	Look for a gallery or upload option on the capture screen	No such option exists	FR-FLD-007	BR-19
TC-18	Installation inside the geofence	Capture both photographs at the site point	Evidence Valid; installation Complete; item Installed; site report updated	FR-EXC-001–002; FR-FLD-003	—
TC-19	GPS mismatch	Capture the AFTER photograph 3 km from the site	Location Exception; Pending Review; not complete; Monitoring Officer notified	FR-EXC-002–003	BR-06
TC-20	Review with a short justification, and self-review	Approve with “ok”; attempt review as the capturing officer	Refused — VAL-EXC-001; refused — VAL-EXC-002	FR-EXC-006, 008	BR-09, BR-27
TC-21	Approve an exception	Approve with a written justification	Installation Complete; item Installed; justification in RPT-10	FR-EXC-009	BR-09

ID	Scenario	Steps	Expected result	Requirements	Rules
TC-22	Offline capture and duplicate synchronisation	Capture offline; reconnect; force a second sync of the same transaction	One installation only; original capture times and coordinates kept; sync time recorded	FR-FLD-012–013	BR-29, BR-30
TC-23	Report and return an out-of-order item	Report failure at site; receive at store	Item in Quarantine; not available for issue	FR-INV-030–031	BR-07, BR-22
TC-24	Classify a return as Usable	Classify Usable and scan to a Stock bin	Item In Store and available	FR-INV-032–033	—
TC-25	Warranty claim and re-inward	Receive a failure within warranty; route to claim; re-inward	Warranty computed Yes; claim recorded; same QR back in stock with full history	FR-INV-034–036	BR-23
TC-26	Scrap and disposal	Try to dispose a Repairable item; classify Not repairable; approve disposal	First refused — VAL-INV-033; after approval item Disposed and in RPT-13	FR-INV-037	BR-24
TC-27	Out-of-scope access	As a Zone North user, open and edit a Zone South requisition	Refused — VAL-SEC-001; attempt audited	FR-SEC-002–003	—
TC-28	Trace an item	Search an installed QR	One view shows PO → GRN → bin → requisition → issue → evidence → status	FR-RPT-006	—
TC-29	Stock reconciliation	Run RPT-04 for a period with receipts, issues, returns and an adjustment	Opening + Receipts + Returns − Issues − Adjustments = Closing for every line	FR-INV-016	BR-08
TC-30	Audit completeness	Perform one action of each type; open the audit trail	Every action shows user, role, time, address and before / after values	FR-SEC-005–006	BR-26

22. Open Issues and Decisions Required

ID	Decision	Owner	Needed by
D-01	Confirm the location hierarchy Store / Warehouse → Rack → Shelf → Bin, and that Shelf may be blank for yard stock.	Store / Warehouse Officer (lead)	Before build of MDM
D-02	Confirm the Project / Monitoring Officer as the exception reviewer, with Lighting Supervisor field verification on request.	Project Director	Before build of EXC
D-03	Define the approval matrix: value bands, approving levels and escalation periods.	Project Director	Before configuration
D-04	Confirm who approves an over-receipt exception — Approving Officer (proposed) or Procurement / Admin.	Procurement / Admin	Before build of INV
D-05	Confirm geofence and tolerance defaults by site type (road stretch, junction, single pole).	Project / Monitoring Officer	Before field pilot
D-06	Decide whether photographs are mandatory for out-of-order returns.	Store / Warehouse Officer	Before build of returns
D-07	Decide the field device policy (official devices or own devices) and the offline retention window.	System Administrator	Before field pilot
D-08	Confirm Procurement / Admin as owner of warranty claims with vendors.	Procurement / Admin	Before go-live
D-09	Confirm retention periods for transactions and evidence under the records policy.	Records / Legal	Before go-live
D-10	Confirm requisition priority levels and whether Urgent / Emergency follow a different approval path.	Approving Officer	Before configuration

Annexure A — Gap Analysis Against the SLIMS Reference Build

A working reference build of SLIMS exists (portal and API, version 0.1). This annexure compares it with the FRD so that the remaining work is visible. It is informative; the requirements in Sections 5 to 21 govern.

Already built and verified in the reference implementation: QR generation with check value and PO-referenced item codes; label printing; scan validation at receipt and issue; requisition, approval with reservation and Issue Slip; scan-based issue with gate pass and custody transfer to the site; field acknowledgement; geo-tagged installation with geofence validation; exception approval with written justification; segregation-of-duty checks on approvals and reviews; quarantine returns with computed warranty; repair re-inward on the same QR; approved disposal; scope enforcement at the service interface; append-only audit trail; fifteen reports; and a dashboard. An automated end-to-end check of 68 assertions passes against the current build.

Area	FRD requirement	Reference build today	Change needed
Location hierarchy	FR-MDM-012	Store ▸ Section ▸ Rack ▸ Bin	Rename Section to Rack and Rack to Shelf, or add Shelf; regenerate location codes
Receiving location and separate put-away	FR-INV-010, FR-INV-011	Items are binned when the GRN is posted	Add Receiving location type and a put-away function; keep one-step receipt as an option
Condition and per-item rejection at receipt	FR-INV-005	Remarks only; discrepancy notes raised separately	Add condition field and per-item rejection linked to a discrepancy note
Over-receipt exception	FR-INV-004	Configurable tolerance percentage only	Add exception request and approval
Roles	Section 4	Store In-Charge, Store Keeper, Supervisor, Officer, PD Office, Procurement, Audit, Administrator	Add Approving Officer, Project / Monitoring Officer and Management / Dashboard User; migrate accounts
Approval authority	FR-REQ-010	Single approve permission, no value limit	Add approval matrix with value bands and escalation
Requisition priority	FR-REQ-001	Purpose and need-by date only	Add priority; make justification mandatory for Urgent / Emergency
Reservation lapse	FR-REQ-014	Parameter exists; nothing releases reservations	Add scheduled release and notification
Exception reviewer	FR-EXC-005	Resolved by any role with verify permission (Supervisor, PD Office)	Assign to Project / Monitoring Officer; add field-verification request
Evidence status values	FR-EXC-002	Confirmed / Pending Verification / Approved Exception / Rejected	Adopt Valid / Location Exception / Missing Evidence with a separate review status
Validation uses both photographs	FR-EXC-001	Distance measured from the AFTER photograph only	Measure both and use the larger
Original photograph preserved	FR-FLD-009	Stamp burned in on the device; original discarded; no hash	Upload original plus metadata; generate stamped copy on the server; store SHA-256
Offline capture	FR-FLD-012	Capture requires connectivity	Deliver field screens as an offline-capable app with a sync queue
Idempotency	FR-FLD-013, FR-SEC-007	Not implemented	Add client transaction IDs and duplicate rejection on every posting
Mock-location detection	FR-FLD-011	Accuracy and clock-drift checks only; a browser cannot see mock providers	Requires the native app
Usable return route	FR-INV-033	Repairable / Warranty / Scrap only (unused returns handled separately)	Add Usable classification
Transfers, adjustments, physical verification	FR-INV-019, FR-INV-020, FR-INV-021	Not implemented	Build
Stock reconciliation report	FR-INV-016	Not implemented	Build RPT-04
Amendment mechanism	FR-EXC-011	GRN reversal only	Build amendment request and approval

Area	FRD requirement	Reference build today	Change needed
Notifications and alerts	Section 14	Not implemented; the dashboard lists pending work	Build notification engine and SMS / e-mail gateway
Export	FR-RPT-007	CSV export and print	Add Excel and PDF export with filters
Map view	FR-RPT-009	Not implemented	Build (Should have)
Minimum GPS accuracy default	Section 12.5	25 m	Set to 20 m

Annexure B — Glossary

Term	Meaning
Approving Officer	Officer who approves requisitions and other controlled actions within delegated authority
Article	A type of material bought against one PO line, identified by its item code
Bin	The lowest addressable store position (Store → Rack → Shelf → Bin)
Evidence status	The system's verdict on installation evidence: Valid, Location Exception or Missing Evidence
Geofence	The radius around a site's recorded point within which installation evidence must fall
GRN	Goods Receipt Note — the record of material received against a PO
Idempotency	The property that posting the same transaction twice has the effect of posting it once
Issue Slip / Gate Pass	Documents generated on approval and on issue respectively
Lighting Officer / Supervisor	The field installer, and the officer who plans site work and raises requisitions
Project / Monitoring Officer	Officer who monitors progress, validates evidence and reviews exceptions
Put-away	Moving received material from Receiving to a Stock bin, making it available
QR identifier	The unique code printed on each label, identifying one serialised unit or one pack
Quarantine	Location holding returned material until it is inspected and classified
Reservation	Stock committed to an approved requisition, unavailable to others
Review status	The state of an exception: Pending Review, Approved, Re-capture Required or Rejected
Site ID	The unique identifier of a lighting site in the site master
Stamped copy	System-generated copy of an evidence photograph with date, time, coordinates and identifiers overlaid
Tolerance	Allowance added to the geofence radius for GPS drift